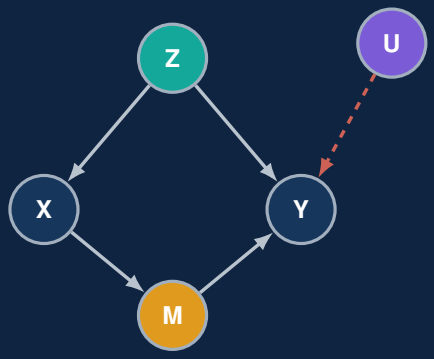


ADVANCED ARTIFICIAL INTELLIGENCE

Probabilistic Graphical Models and Causal Inference

Session 8

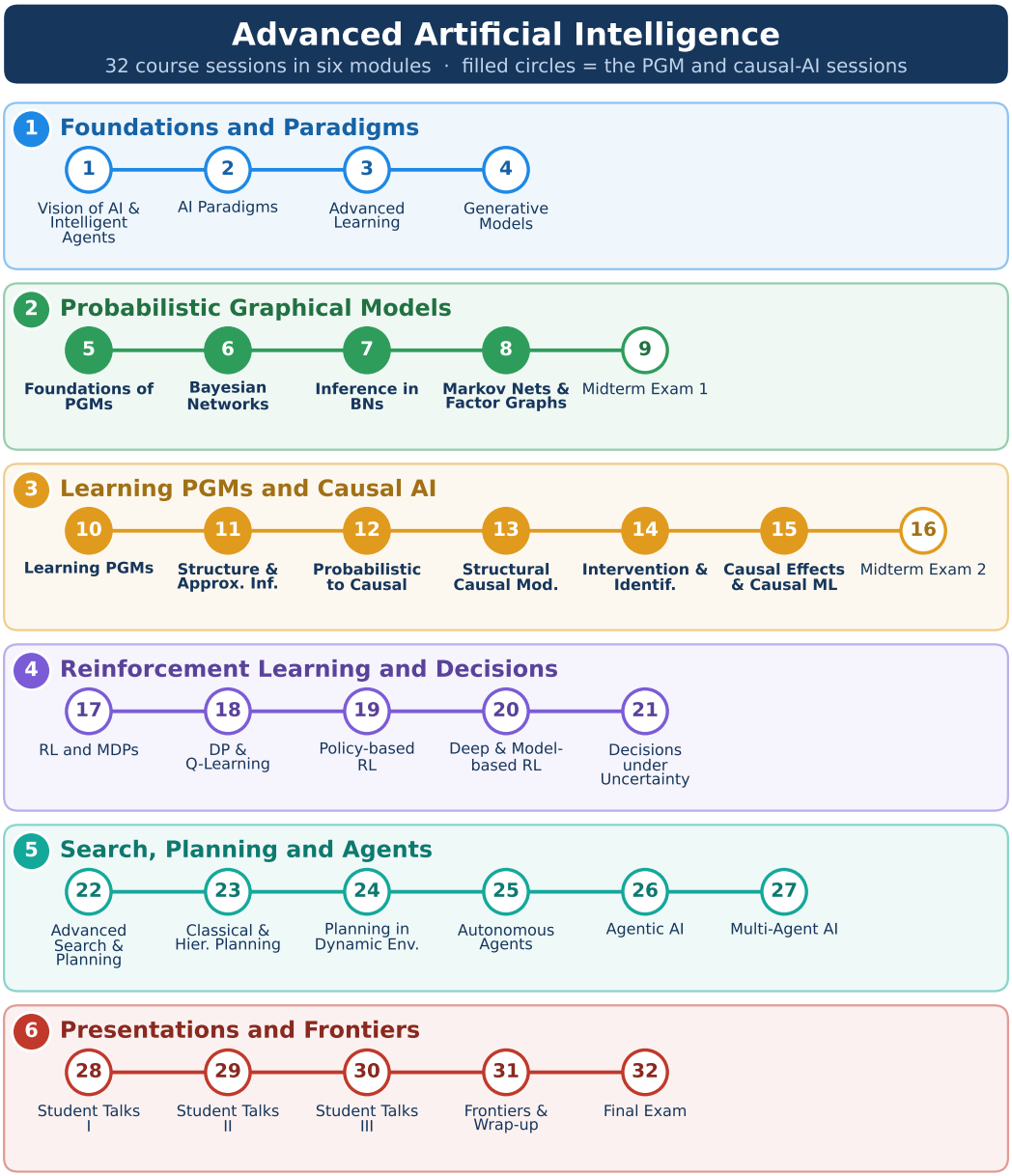
Markov Networks: Undirected Models and Factor Graphs

Lecture Notes

Instructor: Prof. M. Arashi

About these notes

These notes cover Session 8 of the course *Advanced Artificial Intelligence*, part of its probabilistic-graphical-model and causal-AI block (course sessions 5–15). The session follows a fixed structure: a roadmap and learning objectives, a "you are here" course map, the core theory organized in parts, a dedicated part on mathematical statistics, fully worked numerical and analytical examples, and a part connecting the topic to intelligent agents, agentic AI, reinforcement learning, multi-agent systems and the other sessions of the course. Every number is reproduced by the companion Python scripts (`python/s08_examples.py` for the numbers, `python/s08_figures.py` for the figures), and `questions/s08_questions.tex` contains ten questions with complete answers.



The Advanced Artificial Intelligence course. Filled circles mark the PGM and causal-AI sessions.

Contents

Session 8 Markov Networks: Undirected Models and Factor Graphs **1**

8.1 Gibbs Distributions . . . 3
8.2 Reading Independence from an Undirected Graph . . . 4
8.3 Bayesian Networks and Markov Networks Are Incomparable . . . 6
8.4 Log-Linear Models: Ising and Potts . . . 8
8.5 Conditional Random Fields . . . 10
8.6 Factor Graphs . . . 12
8.7 Sum-Product on Factor Graphs . . . 13
8.8 Loopy Belief Propagation . . . 15
8.9 Exponential Families and the Log-Partition Function . . . 16
8.10 Maximum Likelihood is Moment Matching . . . 17
8.11 Pseudo-Likelihood . . . 19
8.12 Worked Examples . . . 21
8.13 Energy-Based Generative Models . . . 23
8.14 Coordination Graphs: Multi-Agent Max-Sum . . . 25
8.15 Summary . . . 28
8.16 Appendix: Proof of the Hammersley–Clifford Theorem . . . 29

SESSION 8 ADVANCED ARTIFICIAL INTELLIGENCE

Markov Networks: Undirected Models and Factor Graphs

Undirected graphical models, Markov random fields, factor graphs and message passing

Advanced Artificial Intelligence · Probabilistic Graphical Models and Causal Inference

Instructor: Prof. M. Arashi

PART I
Undirected Models
Gibbs distributions, separation, Markov properties, Hammersley–Clifford, BN vs MN.

PART II
Markov Random Fields
Log-linear models, Ising and Potts, Gaussian MRFs, conditional random fields.

PART III
Factor Graphs
Bipartite factorizations, sum-product and max-sum messages, loopy BP.

PART IV
Mathematical Statistics
Exponential families, moment matching, Gaussian MRFs, pseudo-likelihood.

PART V
AI Connections
Energy-based models, perception, CRF tools, coordination graphs, factored RL.

LEARNING OBJECTIVES

- Write a Gibbs distribution and its partition function Z (the computational bottleneck); read conditional independences by graph separation; state and prove the relations between the global, local and pairwise Markov properties, and state the Hammersley–Clifford theorem with its positivity condition.
- Moralize a DAG and explain, with counterexamples, why Bayesian and Markov networks are incomparable families.
- Parameterize Markov random fields as log-linear (exponential-family) models: Ising, Potts, Gaussian MRFs and conditional random fields.
- Build factor graphs from BNs and MNs, run sum-product and max-sum message passing by hand, and explain what loopy BP computes.
- Prove $\nabla A = \mathbb{E}[f]$, $\nabla^2 A = \mathrm{Cov}[f]$, derive maximum likelihood as moment matching, and compare it with Besag's pseudo-likelihood.
- Recognize undirected models in energy-based generative models, robot perception, LLM-agent tool pipelines, multi-agent coordination and factored RL.

Where we are. Session 5 introduced conditional independence and factorization, Session 6 encoded them in DAGs, and Session 7 computed with the resulting factors by variable elimination and belief propagation. Today we drop the arrows: undirected models describe symmetric interactions,

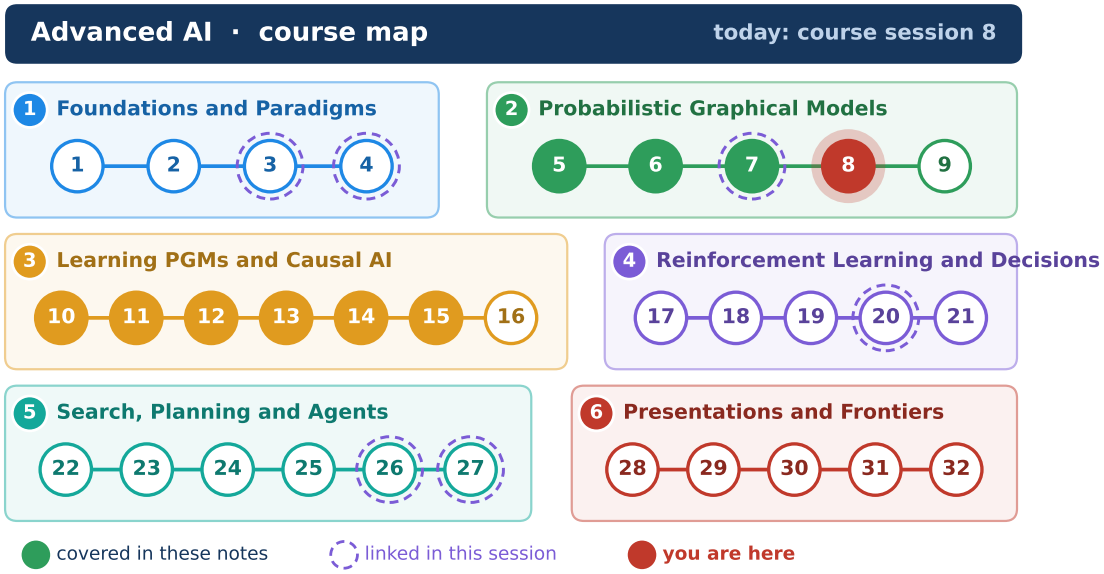


Figure 8.1. Where this session sits in the Advanced AI course (course session 8). Dashed purple rings mark the sessions this chapter links to explicitly.

and factor graphs give one picture in which Session 7's message passing runs unchanged. Session 10 learns parameters and Session 11 learns structure and samples; energy-based models (course session 4), multi-agent coordination (course session 27) and factored RL (course session 20) build on today's material.

OPENING MOTIVATION

Consider modeling a photograph as a grid of pixels, where neighboring pixels tend to have similar brightness. Or a social network, where friends' opinions tend to align. In both cases there is a clear dependency structure, but no natural direction: pixel (i, j) does not "cause" pixel $(i, j+1)$'s brightness any more than the reverse, and my opinion influencing my friend's is just as true as the reverse. Forcing a DAG onto such symmetric relationships means arbitrarily picking an ordering and inventing a fictitious generative story — and, worse, as we will see today, some independence patterns literally cannot be represented by *any* DAG. Markov networks (also called Markov random fields) drop the directionality requirement entirely and represent dependency with plain undirected edges.

Robot perception

An occupancy map from a noisy lidar is a grid of binary cells; an Ising prior "neighbouring cells agree" removes speckle noise (Part V).

Language and LLM agents

A conditional random field tags the tokens of a tool's output (city, date, price) jointly, so that adjacent labels are consistent.

Teams of agents

Robots, drones or software agents whose payoffs depend on their neighbours' actions maximize $\sum_{(i,j)} Q_{ij}(a_i, a_j)$ by exchanging messages.

PART I

Undirected Graphical Models

Potentials instead of conditional probabilities; separation instead of d-separation

PART I · UNDIRECTED MODELS

8.1 Gibbs Distributions

Definition 8.1 (Gibbs distribution / Markov network)**.** Given an undirected graph H over variables $X_1, \dots, X_n$, a set of *potential functions* (or factors) $\phi_1, \dots, \phi_m$, one per clique C_k (a fully-connected subset) of H, defines a joint distribution

$$P(x_1, \dots, x_n) = \frac{1}{Z} \prod_{k=1}^{m} \phi_k(x_{C_k}), \qquad Z = \sum_{x_1, \dots, x_n} \prod_{k=1}^{m} \phi_k(x_{C_k}), \tag{8.1}$$

where Z, the *partition function*, is a normalizing constant.

Unlike a Bayesian network's CPTs, a potential ϕ_k need not sum to one and need not represent "a probability of something given something else" — it is simply a non-negative compatibility score: how much the joint configuration x_{C_k} is favored, in relative terms. This is precisely the generalization hinted at by Session 7's recommendation-system factor graph (factors $\phi_1(C), \phi_2(C, X), \phi_3(C, X, Y)$) and by its pairwise drone-line potentials: factors as raw compatibility scores, not necessarily conditional probabilities.

Since a potential on a sub-clique can always be multiplied into a potential on any clique containing it, we may assume without loss of generality that the C_k are the *maximal* cliques of H. The most common special case is the *pairwise Markov network*, with unary and edge potentials only,

$$P(\mathbf{x}) = \frac{1}{Z} \prod_{i \in V} \phi_i(x_i) \prod_{(i,j) \in \mathcal{E}} \psi_{ij}(x_i, x_j). \tag{8.2}$$

For strictly positive potentials we write $\phi_k = e^{-E_k}$ and obtain the *energy form* $P(\mathbf{x}) = \exp\{-E(\mathbf{x})\}/Z$ with $E = \sum_k E_k$: low energy means high probability.

INTUITION

Think of $E_k = -\log \phi_k$ as an "energy" contribution: configurations with high potential (low energy) are more probable, configurations with low potential are penalized, and Z simply rescales everything to sum to one. In an image-denoising grid (Fig. 8.2), a pairwise potential $\psi_{ij}(x_i, x_j)$ that is large when $x_i = x_j$ and small otherwise encodes "neighboring pixels prefer to agree" — with no claim about which pixel came first.

The crucial computational fact: *variable elimination from Session 7 applies to Gibbs distributions with no modification at all.* VE only ever needed factors that could be multiplied and summed — it never used the fact that BN factors happened to be conditional probability tables. This is the payoff promised at the end of Session 7. The one new ingredient is that the final sum is no longer 1: eliminating *all* variables returns Z itself (Worked Example 8.13).

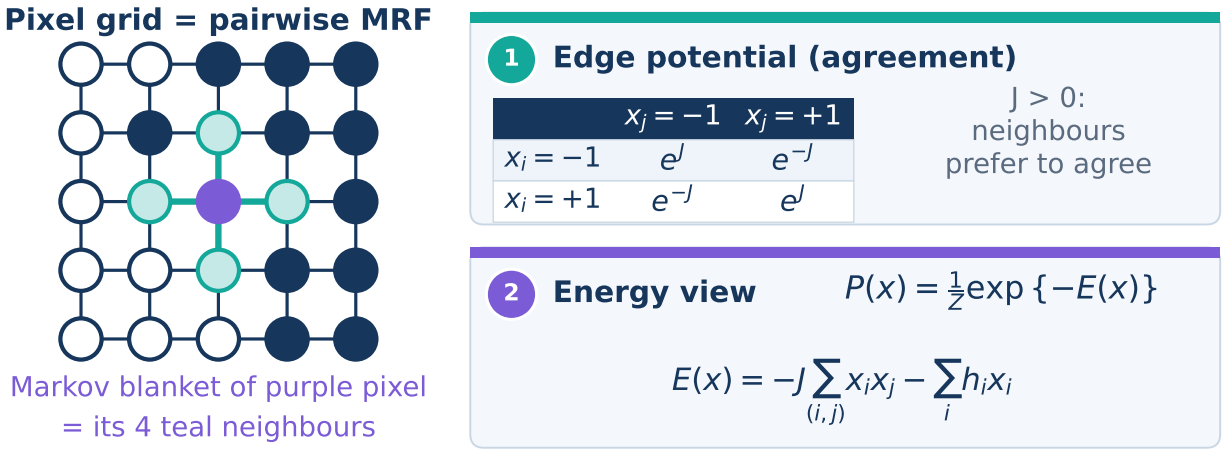


Figure 8.2. A pairwise Markov network on a pixel grid. Every edge carries a potential favoring agreement between neighbors; there is no natural direction to any edge. Conditioning on its four neighbours shields a pixel from the rest of the image.

PART I · UNDIRECTED MODELS

8.2 Reading Independence from an Undirected Graph

Markov networks have their own, and in fact *simpler*, independence-reading rule than d-separation — no colliders to worry about.

Definition 8.2 (Global Markov property)**.** Let A, B, S be disjoint sets of nodes of the undirected graph H. A and B are *separated* by S, written $A \perp_H B \mid S$, if every path between a node of A and a node of B passes through some node in S (ordinary graph separation — delete S and check whether A and B land in different connected components). P satisfies the *global Markov property* (G) w.r.t. H if $A \perp_H B \mid S$ implies $\mathbf{X}_A \perp\!\!\!\perp \mathbf{X}_B \mid \mathbf{X}_S$ for all disjoint A, B, S. (We write S, not Z, for separators: Z is reserved for the partition function. Below we write $A \perp\!\!\!\perp B \mid S$ for $\mathbf{X}_A \perp\!\!\!\perp \mathbf{X}_B \mid \mathbf{X}_S$.)

Definition 8.3 (Local and pairwise Markov properties)**.** Let $N(i)$ be the neighbours of node i. P satisfies the *local* property (L) if $X_i \perp\!\!\!\perp X_{V\setminus(\{i\}\cup N(i))} \mid X_{N(i)}$ for every i, and the *pairwise* property (P) if $X_i \perp\!\!\!\perp X_j \mid X_{V\setminus\{i,j\}}$ for every non-adjacent pair i, j. In a Markov network the Markov blanket of a node is simply its set of neighbours.

Proposition 8.1 (Gibbs ⇒ global Markov)**.** *If P factorizes as in (8.1) over the cliques of H, then P satisfies (G). No positivity is needed.*

Proof. Let S separate A from B. Let $\tilde{A}$ be the set of nodes connected to A in the graph with S removed ($A \subseteq \tilde{A}$), and $\tilde{B} = V \setminus (\tilde{A} \cup S) \supseteq B$. There is no edge between $\tilde{A}$ and $\tilde{B}$, so every clique lies inside $\tilde{A} \cup S$ or inside $\tilde{B} \cup S$. Grouping the potentials accordingly, $P(\mathbf{x}) = g(x_{\tilde{A}}, x_S)\, h(x_{\tilde{B}}, x_S)$, which is the factorization criterion for $\tilde{A} \perp\!\!\!\perp \tilde{B} \mid S$ (Session 5). Decomposition then gives $A \perp\!\!\!\perp B \mid S$. □

Proposition 8.2 (The Markov hierarchy)**.** *For every distribution, (G) ⇒ (L) ⇒ (P). If P is strictly positive, (P) ⇒ (G), so the three properties coincide.*

Proof. (G)⇒(L): $N(i)$ separates i from every non-neighbour. (L)⇒(P): for non-adjacent j, (L) gives $X_i \perp\!\!\!\perp X_{V\setminus \mathrm{cl}(i)} \mid X_{N(i)}$ with $j \in V \setminus \mathrm{cl}(i)$, $\mathrm{cl}(i) = \{i\} \cup N(i)$; weak union moves the other non-neighbours into the conditioning set.

(P)⇒(G) under positivity, by backward induction on $|S|$. Let S separate A from B, $n = |V|$. If $|S| = n-2$, then A, B are non-adjacent singletons and (P) is exactly the claim. Otherwise, suppose the claim holds for all larger separators. *Case* $A \cup B \cup S = V$*:* one of A, B, say A, has at least two elements; pick $\alpha \in A$. Then $S \cup \{\alpha\}$ separates $A \setminus \{\alpha\}$ from B, and $S \cup (A \setminus \{\alpha\})$ separates α from B, so by induction $A \setminus \{\alpha\} \perp\!\!\!\perp B \mid S \cup \{\alpha\}$ and $\alpha \perp\!\!\!\perp B \mid S \cup A \setminus \{\alpha\}$; the intersection property (valid for positive P, Session 5) gives $A \perp\!\!\!\perp B \mid S$. *Case* $A \cup B \cup S \neq V$*:* pick $\gamma \notin A \cup B \cup S$. $S \cup \{\gamma\}$ separates A from B, so $A \perp\!\!\!\perp B \mid S \cup \{\gamma\}$. Moreover γ cannot reach both A and B avoiding S, so, say, $A \cup S$ separates γ from B and $\gamma \perp\!\!\!\perp B \mid A \cup S$. Intersection gives $A \cup \{\gamma\} \perp\!\!\!\perp B \mid S$, and decomposition $A \perp\!\!\!\perp B \mid S$. □

Worked Example 8.1 (Pairwise does not imply global without positivity)**.** Let $X_1 = X_2 = X_3 \sim \mathrm{Bernoulli}(\frac{1}{2})$ (three perfect copies, so $P(0,0,0) = P(1,1,1) = \frac{1}{2}$ and all other six configurations have probability 0) and let H be the *empty* graph on $\{X_1, X_2, X_3\}$. In the empty graph every pair is non-adjacent and $V \setminus \{i,j\} = \{k\}$, so (P) requires $X_i \perp\!\!\!\perp X_j \mid X_k$ for the three pairs. Given $X_k = x_k$ the other two variables are the constants x_k, and constants are independent of everything, so (P) holds. For (L) and (G), however, $N(i) = \emptyset$ and the empty set separates everything, so both require *marginal* independence: but $P(X_1{=}1) = P(X_2{=}1) = \frac{1}{2}$, so $P(X_1{=}1)P(X_2{=}1) = 0.25$, while $P(X_1{=}1, X_2{=}1) = P(1,1,1) = 0.5 \neq 0.25$. Hence (L) and (G) fail. The proof of (P)⇒(G) breaks at the intersection property, which needs positivity and here $P(0,1,\cdot) = 0$.

PYTHON LAB · s08_examples.py · ex_markov_property_counterexamples()

```
X1 indep X2 | X3:  True ...  (all three pairs True)
P(X1=1,X2=1) = 0.50 vs P(X1=1)P(X2=1) = 0.25 -> marginal CI fails
4-cycle example:  X1 indep X3 | X2,X4 :  True; X2 indep X4 | X1,X3 :  True
P(0010) = 0.0
```

Theorem 8.1 (Hammersley–Clifford)**.** *If P is a strictly positive distribution ($P(\mathbf{x}) > 0$ for all $\mathbf{x}$), then P satisfies the global Markov property with respect to H if and only if P factorizes as a Gibbs distribution over the cliques of H.*

We state this landmark result here and prove it in the appendix (Section 8.16): one direction is Proposition 8.1, the other is a genuinely non-trivial argument via Möbius inversion. What matters for practice is simpler and more useful: *once* you have a Gibbs factorization over H's cliques, ordinary graph separation immediately tells you every conditional independence the model implies — no chain/fork/collider bookkeeping required. Positivity is the one condition worth remembering: without it (e.g., a distribution with hard logical constraints setting some configurations to probability exactly zero), the equivalence can fail.

Remark 8.1 (A Markov but non-Gibbs distribution). On the 4-cycle $X_1 - X_2 - X_3 - X_4 - X_1$ put probability $\frac{1}{8}$ on each of $0000, 1000, 1100, 1110, 1111, 0111, 0011, 0001$ (Koller & Friedman 2009, Ch. 4). Both separation statements $X_1 \perp\!\!\!\perp X_3 \mid X_2, X_4$ and $X_2 \perp\!\!\!\perp X_4 \mid X_1, X_3$ hold, so (G) holds. To check the first, fix (x_2, x_4): exactly two support points remain and in each slice one of X_1, X_3 is constant — $(0,0)$ leaves $0000, 1000$ ($x_3 = 0$), $(1,0)$ leaves $1100, 1110$

($x_1 = 1$), $(1,1)$ leaves $1111, 0111$ ($x_3 = 1$), $(0,1)$ leaves $0011, 0001$ ($x_1 = 0$); a constant is independent of anything. The second statement is checked the same way with the slices of (x_1, x_3) (`ex_markov_property_counterexamples()` verifies both). If $P = \phi_{12}\phi_{23}\phi_{34}\phi_{41}/Z$, then every factor evaluated at a support point must be positive (a product is positive only if all its terms are). Hence $\phi_{12}(0,0), \phi_{41}(0,0) > 0$ (from 0000), $\phi_{23}(0,1) > 0$ (from 0011, where $x_2 = 0, x_3 = 1$) and $\phi_{34}(1,0) > 0$ (from 1110, where $x_3 = 1, x_4 = 0$), forcing $P(0010) = \phi_{12}(0,0)\phi_{23}(0,1)\phi_{34}(1,0)\phi_{41}(0,0)/Z > 0$ — but $P(0010) = 0$. Without positivity, (G) does not imply a Gibbs factorization.

PART I · UNDIRECTED MODELS

8.3 Bayesian Networks and Markov Networks Are Incomparable

Every DAG has an undirected "shadow" that captures its independencies, but building it requires one extra step beyond simply dropping arrowheads.

Definition 8.4 (Moralization)**.** The *moral graph* of a DAG G is the undirected graph obtained by (1) connecting every pair of parents that share a common child ("marrying" them), and (2) dropping all edge directions.

Proposition 8.3. *Every distribution representable by a Bayesian network with DAG G is representable by a Markov network on the moral graph of G (set each clique potential equal to the corresponding CPT). The converse is not generally true.*

Proof. In the moral graph each family $\{X_i\} \cup \mathrm{pa}(X_i)$ is a clique: every parent is joined to X_i, and the parents are married to each other. So $\phi_i(x_i, x_{\mathrm{pa}(i)}) := P(x_i \mid x_{\mathrm{pa}(i)})$ is a clique potential and $\prod_i \phi_i = P$ with $Z = 1$. For the converse see the 4-cycle below. □

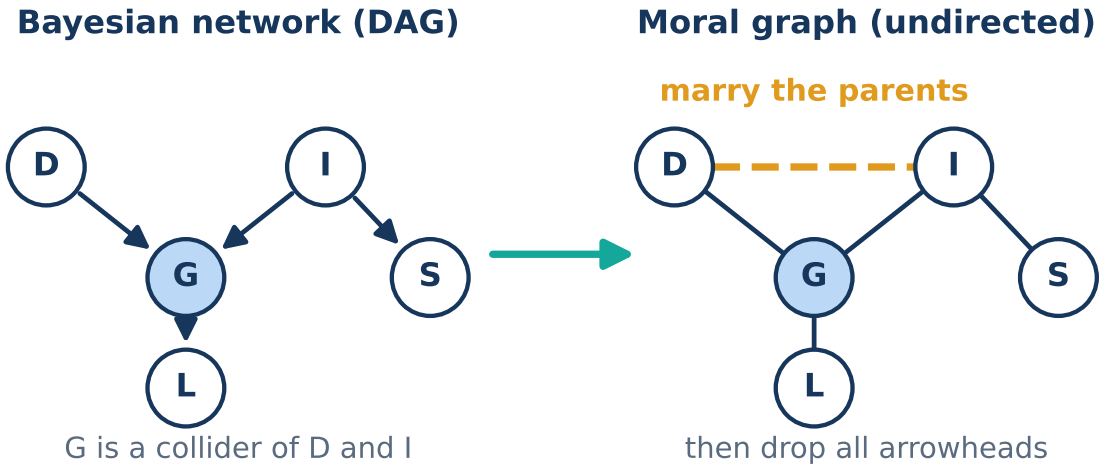


Figure 8.3. Moralizing the student network: parents D, I of the collider G are married with a new edge before dropping directions. Without this edge, ordinary separation would wrongly claim $D \perp I \mid G$, when actually D and I become dependent once G is observed (Session 6's explaining-away).

The marriage step is not bookkeeping trivia — it is forced by exactly the collider asymmetry from Session 6. If we only dropped the arrowheads, D and I would be joined only through the path D–G–I, so ordinary separation would claim $D \perp I \mid G$ — false, because observing the collider G makes D and I dependent (explaining-away; Worked Example 8.15 computes $P(D{=}1, I{=}1 \mid G{=}1) = 0.3333$ versus 0.4214 under the false independence). Marrying D and I removes this wrong claim. The price is that the moral graph can no longer express the true marginal independence $D \perp\!\!\!\perp I$, since D–I is now an edge.

But moralization is a one-way translation, and the families are genuinely incomparable:

- A collider $X \to Z \leftarrow Y$ encodes $X \perp\!\!\!\perp Y$ (marginally) but $X \not\perp\!\!\!\perp Y \mid Z$. No undirected graph on three nodes can express *both* facts with the separation rule, because separation is monotone in the conditioning set: if no path joins X and Y (so $X \perp_H Y \mid \emptyset$), then no path joins them after deleting Z either, so $X \perp_H Y \mid Z$ is also asserted. To avoid asserting $X \perp_H Y \mid Z$ one needs a path from X to Y avoiding Z, i.e. the edge X–Y, which in turn destroys $X \perp_H Y$.
- Conversely, a 4-cycle A–B–C–D–A with no chord encodes $A \perp\!\!\!\perp C \mid \{B, D\}$ and $B \perp\!\!\!\perp D \mid \{A, C\}$ simultaneously. No DAG on four nodes reproduces exactly this independence pattern (any acyclic orientation of the cycle creates at least one collider, breaking the required symmetry; Worked Example 8.12).

Figure 8.4 summarizes this: the two model classes overlap (e.g., trees and chains can be written either way) but neither contains the other. The overlap is exactly characterized: an independence model has both a DAG and an undirected perfect map if and only if it has a *chordal* (triangulated, decomposable: every cycle of length ≥ 4 has a chord) undirected perfect map (Koller & Friedman 2009, Ch. 4) — the graphs on which Session 7's junction trees are built.

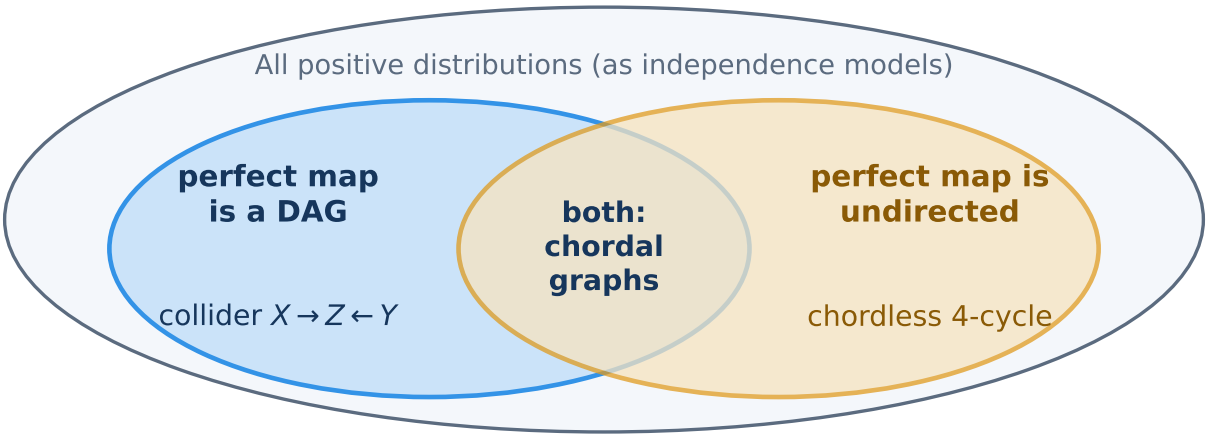


Figure 8.4. Bayesian networks and Markov networks are complementary, not nested, families of representable independence structures; their intersection is the chordal graphs. Outside both lie, e.g., models combining a collider with a chordless 4-cycle.

PART II

Log-Linear Models and Markov Random Fields

Ising, Potts, Gaussian fields and conditional random fields

Table 8.1. Directed versus undirected graphical models at a glance.

	Bayesian network (DAG)	Markov network (undirected)
Local parameters	CPTs $P(x_i \mid \mathrm{pa}_i)$, normalized	potentials $\phi_C \geq 0$, unnormalized
Normalizer	$Z = 1$ automatically	$Z = \sum_{\mathbf{x}} \prod_C \phi_C$, often intractable
Independence rule	d-separation (colliders!)	plain separation
Natural for	causal / generative stories	symmetric interactions, constraints
Sampling	ancestral, exact and cheap	MCMC (Gibbs, Session 11)
Learning (complete data)	closed form, counts (Session 10)	iterative; each step needs inference
Cannot express	chordless 4-cycle	collider: $X \perp\!\!\!\perp Y$ but $X \not\perp\!\!\!\perp Y \mid Z$

PART II · RANDOM FIELDS

8.4 Log-Linear Models: Ising and Potts

Every strictly positive Gibbs distribution can be rewritten with features f_k and weights θ_k:

$$P_\theta(\mathbf{x}) = \exp\Big\{\sum_{k=1}^{K} \theta_k f_k(\mathbf{x}_{C_k}) - A(\theta)\Big\}, \qquad A(\theta) = \log \sum_{\mathbf{x}} \exp\Big\{\sum_{k} \theta_k f_k(\mathbf{x}_{C_k})\Big\} = \log Z(\theta). \quad (8.3)$$

Taking indicator features $f_{C,\mathbf{u}}(\mathbf{x}) = \mathbb{1}[\mathbf{x}_C = \mathbf{u}]$ and $\theta_{C,\mathbf{u}} = \log \phi_C(\mathbf{u})$ recovers any table potential, but the strength of (8.3) is that a few well-chosen features suffice. This is an *exponential family* with sufficient statistics f and log-partition (cumulant) function A; Part IV studies it.

Definition 8.5 (Ising and Potts models)**.** The *Ising model* on a graph $(V, \mathcal{E})$ has spins $x_i \in \{-1, +1\}$ and

$$P(\mathbf{x}) = \frac{1}{Z} \exp\Big\{\sum_{(i,j)\in\mathcal{E}} J_{ij} x_i x_j + \sum_i h_i x_i\Big\}, \quad (8.4)$$

with couplings J_{ij} (ferromagnetic if $J_{ij} > 0$: neighbours prefer to agree) and fields h_i. The *Potts model* allows q states and rewards agreement: $P(\mathbf{x}) \propto \exp\{\sum_{(i,j)} J_{ij} \mathbb{1}[x_i = x_j] + \sum_i h_i(x_i)\}$; $q = 2$ is the Ising model up to reparametrization.

Worked Example 8.2 (Two spins, analytically)**.** For two spins with $P(x_1, x_2) \propto e^{J x_1 x_2}$, find Z, $\mathbb{E}[x_1 x_2]$ and $\mathrm{Var}[x_1 x_2]$, and verify $A'(J) = \mathbb{E}[x_1 x_2]$, $A''(J) = \mathrm{Var}[x_1 x_2]$.

Solution. The four configurations have $x_1 x_2 = +1$ (for $(+1,+1), (-1,-1)$) or -1 (for $(+1,-1), (-1,+1)$), twice each, so $Z = 2e^{J} + 2e^{-J} = 4\cosh J$ and $A(J) = \log(4\cosh J)$. Then $\mathbb{E}[x_1 x_2] = (2e^{J} - 2e^{-J})/Z = \tanh J = A'(J)$ and, since $(x_1 x_2)^2 = 1$, $\mathrm{Var}[x_1 x_2] = \mathbb{E}[(x_1 x_2)^2] - (\mathbb{E}[x_1 x_2])^2 = 1 - \tanh^2 J = A''(J)$ (because $\frac{d}{dJ}\tanh J = 1 - \tanh^2 J$). Numerically, at $J = 0.5$:

$$\begin{aligned} Z &= 2e^{0.5} + 2e^{-0.5} = 2(1.6487) + 2(0.6065) = 4\cosh 0.5 = 4(1.1276) = 4.5105, \\ A(0.5) &= \log 4.5105 = 1.5064, \qquad \mathbb{E}[x_1 x_2] = \tanh 0.5 = 0.4621, \\ \mathrm{Var}[x_1 x_2] &= 1 - 0.4621^2 = 1 - 0.2136 = 0.7864. \end{aligned}$$

Central finite differences of A (step 10^{-4}) reproduce $A'(0.5) = 0.4621$ and $A''(0.5) = 0.7864$. So A is convex, its slope is the mean of the sufficient statistic and its curvature the variance (the Fisher information) — the general pattern of Theorem 8.3.

PYTHON LAB · s08_examples.py · ex_ising_two_spin()

```
e^J = 1.6487, e^-J = 0.6065, cosh J = 1.1276
Z = 4 cosh J = 4.5105, A(J) = log Z = 1.5064
E[x1 x2] = tanh J = 0.4621; tanh^2 J = 0.2136; Var[x1 x2] = 0.7864
finite-difference A'(J) = 0.4621, A''(J) = 0.7864
```

Worked Example 8.3 (The conditional of one spin is logistic)**.** Show that in (8.4) $P(x_i{=}{+}1 \mid \mathbf{x}_{-i}) = \sigma\big(2(h_i + \sum_{j\in N(i)} J_{ij}x_j)\big)$, $\sigma(t) = 1/(1+e^{-t})$.
Solution. The terms of the exponent that contain x_i are $x_i\, m_i$ with the *local field* $m_i = h_i + \sum_{j\in N(i)} J_{ij}x_j$. In the ratio $P(x_i{=}+1,\mathbf{x}_{-i})/P(x_i{=}-1,\mathbf{x}_{-i})$ the partition function and every term not containing x_i cancel, leaving $e^{(+1)m_i}/e^{(-1)m_i} = e^{2m_i}$. A probability q with odds $q/(1-q) = e^t$ is $q = e^t/(1+e^t) = \sigma(t)$, so $P(x_i{=}+1 \mid \mathbf{x}_{-i}) = \sigma(2m_i)$. Only the neighbours enter — the local Markov property in formula form. Numerically, with four neighbours $(+1,+1,-1,+1)$, all couplings $J_{ij} = 0.5$ and $h_i = 0.1$: $m_i = 0.1 + 0.5(1+1-1+1) = 0.1 + 1.0 = 1.10$, and $P(x_i{=}{+}1 \mid \cdot) = \sigma(2.2) = 1/(1+e^{-2.2}) = 1/(1+0.1108) = 0.9002$. This one line is the Gibbs sampler of the Ising model (Session 11) and the reason pseudo-likelihood is a set of logistic regressions (Section 8.11).

PYTHON LAB · s08_examples.py · ex_ising_conditional()

```
neighbours = (1, 1, -1, 1), J = 0.5, h = 0.1:  local field = 1.10
e^(-2*1.10) = 0.1108
P(x_i=+1 | rest) = sigma(2*1.10) = 0.9002
```

Phase transition (simulation). On the infinite square lattice ($h = 0$, equal couplings J) the Ising model is disordered for $J < J_c$ and magnetized for $J > J_c = \tfrac{1}{2}\ln(1+\sqrt{2}) = \tfrac{1}{2}\ln 2.4142 = 0.4407$ (Onsager). To see this on a finite lattice, `ex_ising_gibbs()` runs the Gibbs sampler of Worked Example 8.3 on a 48×48 lattice with free boundary, $h = 0$, starting from independent random spins (seed 4), for 400 sweeps (each sweep updates the two checkerboard halves in turn). It reports the magnetization $|\frac{1}{48^2}\sum_i x_i|$ averaged over the last 200 sweeps and the fraction of neighbouring pairs that agree in the final sample. These are *simulation outputs*, not exact values: at $J = 0.25, 0.44, 0.70$ the mean |magnetization| is 0.037, 0.219, 0.805 and the neighbour agreement is 0.636, 0.851, 0.981. One local parameter produces qualitatively different global behaviour.

PYTHON LAB · s08_examples.py · ex_ising_gibbs()

```
J = 0.25:  mean |magnetization| (2nd half) = 0.037,
neighbour agreement in final sample = 0.636
J = 0.44:  mean |magnetization| (2nd half) = 0.219,
neighbour agreement in final sample = 0.851
J = 0.70:  mean |magnetization| (2nd half) = 0.805,
neighbour agreement in final sample = 0.981
```

8.4.1 Gaussian Markov random fields

A Gaussian MRF is $\mathbf{x} \sim N(\boldsymbol{\mu}, K^{-1})$ with *precision* matrix $K \succ 0$:

$$p(\mathbf{x}) \propto \exp\Big\{ -\tfrac{1}{2}\mathbf{x}^\top K \mathbf{x} + \mathbf{h}^\top \mathbf{x} \Big\} = \prod_i e^{-\frac{1}{2}K_{ii}x_i^2 + h_i x_i} \prod_{i<j} e^{-K_{ij}x_i x_j}, \qquad \mathbf{h} = K\boldsymbol{\mu}. \tag{8.5}$$

It is a pairwise MRF whose edges are the non-zero off-diagonal entries of K; by Proposition 8.4 (Part IV), $K_{ij} = 0$ is *exactly* $X_i \perp\!\!\!\perp X_j \mid X_{\text{rest}}$, and the local conditionals are $X_i \mid \mathbf{x}_{-i} \sim N\big(\mu_i - \sum_{j\neq i} \frac{K_{ij}}{K_{ii}}(x_j - \mu_j),\ \frac{1}{K_{ii}}\big)$, with partial correlations $\rho_{ij\cdot\text{rest}} = -K_{ij}/\sqrt{K_{ii}K_{jj}}$.

Worked Example 8.4 (Sparse precision, dense covariance)**.** Take the 4-node chain with $K_{ii} = 2$, $K_{i,i+1} = -0.8$ and zeros elsewhere ($\boldsymbol{\mu} = 0$). Compare marginal and partial correlations of X_1, X_4.

Solution. For a tridiagonal matrix the leading principal minors obey $D_k = K_{kk}D_{k-1} - K_{k-1,k}^2 D_{k-2}$ ($D_0 = 1$), here $D_k = 2D_{k-1} - 0.64D_{k-2}$: $D_1 = 2$, $D_2 = 4 - 0.64 = 3.36$, $D_3 = 6.72 - 1.28 = 5.44$, $D_4 = \det K = 10.88 - 2.1504 = 8.7296$. By the cofactor formula $\Sigma = K^{-1} = \operatorname{adj}(K)/\det K$:

$$\Sigma_{14} = \frac{(-1)^{1+4}\det\begin{pmatrix} -0.8 & 2 & -0.8 \\ 0 & -0.8 & 2 \\ 0 & 0 & -0.8 \end{pmatrix}}{\det K} = \frac{0.8^3}{8.7296} = \frac{0.512}{8.7296} = 0.0587,$$

$$\Sigma_{11} = \Sigma_{44} = \frac{\det K_{\{2,3,4\}}}{\det K} = \frac{D_3}{D_4} = \frac{5.44}{8.7296} = 0.6232$$

(the lower-right 3×3 block has the same structure as the upper-left one, so its determinant is also D_3). The marginal correlation is $\operatorname{corr}(X_1, X_4) = \Sigma_{14}/\sqrt{\Sigma_{11}\Sigma_{44}} = 0.512/5.44 = 0.0941 \neq 0$: the ends of the chain are dependent. Yet $K_{14} = 0$, so the partial correlation given X_2, X_3 is $-K_{14}/\sqrt{K_{11}K_{44}} = 0$, while adjacent nodes have partial correlation $-K_{12}/\sqrt{K_{11}K_{22}} = 0.8/\sqrt{2\cdot 2} = 0.4$. The local conditional is $X_1 \mid \text{rest} \sim N\big(-\frac{K_{12}}{K_{11}}x_2,\ \frac{1}{K_{11}}\big) = N(0.4x_2,\ 0.5)$. Covariance zeros would instead encode *marginal* independence.

PYTHON LAB · s08_examples.py · ex_gaussian_mrf()

```
leading determinants D1..D4 = [2.0, 3.36, 5.44, 8.7296]
Sigma_14 = 0.512/8.7296 = 0.0587
Sigma_11 = Sigma_44 = 5.44/8.7296 = 0.6232; corr(X1,X4) = 0.512/5.44 =
0.0941
partial corr(X1,X4 | X2,X3) = 0.0000; adjacent partial corr = 0.4
Var(X1 | rest) = 1/K11 = 0.5000; E[X1 | rest] = 0.4 * x2
```

PART II · RANDOM FIELDS

8.5 Conditional Random Fields

In prediction we observe inputs $\mathbf{x}$ and want labels $\mathbf{y}$; modelling $P(\mathbf{x})$ is wasted effort. A *conditional random field* is a Markov network over $\mathbf{y}$ whose potentials may depend on all of $\mathbf{x}$:

$$P(\mathbf{y} \mid \mathbf{x}) = \frac{1}{Z(\mathbf{x})} \exp\Big\{ \sum_k \theta_k f_k(\mathbf{y}_{C_k}, \mathbf{x}) \Big\}, \qquad Z(\mathbf{x}) = \sum_{\mathbf{y}} \exp\Big\{ \sum_k \theta_k f_k(\mathbf{y}_{C_k}, \mathbf{x}) \Big\}. \tag{8.6}$$

The *linear-chain CRF* has token scores $u_t(y_t;\mathbf{x})$ and transition scores $w(y_{t-1},y_t)$; its graph over $\mathbf{y}$ is a chain, so $Z(\mathbf{x})$, marginals and the best labelling come from Session 7's forward–backward and Viterbi algorithms in $O(TL^2)$. An HMM is the generative counterpart; the discriminative CRF may use features of the whole sentence without modelling the inputs.

Worked Example 8.5 (Tagging a booking request)**.** A hypothetical linear-chain CRF tags "Book Paris hotel" with labels $\{\mathrm{O},\mathrm{LOC}\}$; the score of a labelling is $s(\mathbf{y})=\sum_{t=1}^{3}u_t(y_t)+\sum_{t=2}^{3}w(y_{t-1},y_t)$ and $P(\mathbf{y}\mid\mathbf{x})=e^{s(\mathbf{y})}/Z(\mathbf{x})$. Token log-potentials, listed as (O, LOC): $u_1=(2,0)$, $u_2=(0.5,2)$, $u_3=(1,0.8)$; transitions $w(\mathrm{O},\mathrm{O})=0.5$, $w(\mathrm{LOC},\mathrm{LOC})=1$, $w(\mathrm{O},\mathrm{LOC})=w(\mathrm{LOC},\mathrm{O})=0$. Find $Z(\mathbf{x})$, the marginals and the best labelling.
Solution. (1) *Forward pass* (on the exponential scale), $\alpha_1(y)=e^{u_1(y)}$ and $\alpha_t(y)=e^{u_t(y)}\sum_{y'}\alpha_{t-1}(y')e^{w(y',y)}$; vectors are (O, LOC):

$$\begin{aligned}\alpha_1&=(e^{2},\,e^{0})=(7.3891,\,1),\\ \alpha_2&=\big(e^{0.5}(7.3891e^{0.5}+1),\;e^{2}(7.3891+1\cdot e^{1})\big)=(21.7343,\,74.6837),\\ \alpha_3&=\big(e^{1}(21.7343e^{0.5}+74.6837),\;e^{0.8}(21.7343+74.6837e^{1})\big)=(300.4175,\,500.1805),\end{aligned}$$

so $Z(\mathbf{x})=300.4175+500.1805=800.598$ and $\log Z(\mathbf{x})=6.6854$, identical to summing $e^{s(\mathbf{y})}$ over all 2^3 labellings. (2) *Backward pass*, $\beta_3=(1,1)$, $\beta_t(y)=\sum_{y'}e^{w(y,y')+u_{t+1}(y')}\beta_{t+1}(y')$: $\beta_2=(e^{1.5}+e^{0.8},\,e^{1}+e^{1.8})=(6.7072,\,8.7679)$ and $\beta_1=(e^{1}\cdot 6.7072+e^{2}\cdot 8.7679,\;e^{0.5}\cdot 6.7072+e^{3}\cdot 8.7679)=(83.0189,\,187.1669)$. Marginals $P(y_t{=}\mathrm{LOC}\mid\mathbf{x})=\alpha_t(\mathrm{LOC})\beta_t(\mathrm{LOC})/Z$: $1\cdot 187.1669/800.598=0.2338$ (Book), $74.6837\cdot 8.7679/800.598=0.8179$ (Paris), $500.1805\cdot 1/800.598=0.6248$ (hotel). (3) *Viterbi* (max instead of sum, log scale), $\delta_1=u_1=(2,0)$, $\delta_t(y)=u_t(y)+\max_{y'}[\delta_{t-1}(y')+w(y',y)]$: $\delta_2=(0.5+\max(2.5,0),\,2+\max(2,1))=(3,4)$, both from O; $\delta_3=(1+\max(3.5,4),\,0.8+\max(3,5))=(5,5.8)$, both from LOC. The best final label is LOC (score 5.8); back-tracking gives (O, LOC, LOC), score $u_1(\mathrm{O})+u_2(\mathrm{LOC})+u_3(\mathrm{LOC})+w(\mathrm{O},\mathrm{LOC})+w(\mathrm{LOC},\mathrm{LOC})=2+2+0.8+0+1=5.8$ and $P(\mathbf{y}^\star\mid\mathbf{x})=e^{5.8}/Z=330.30/800.598=0.4126$. "hotel" alone prefers O (u_3: $1>0.8$), but the LOC–LOC transition pulls it into the location phrase: labels are decided *jointly*.

PYTHON LAB · s08_examples.py · ex_crf_chain()

```
alpha = [7.3891, 1.0], [21.7343, 74.6837], [300.4175, 500.1805]
beta_1 = [83.0189, 187.1669], beta_2 = [6.7072, 8.7679]
Viterbi delta = [2, 0], [3, 4], [5, 5.8]
log Z(x) forward = 6.6854, brute force = 6.6854, Z(x) = 800.598
P(y_t=LOC | x) = 0.2338, 0.8179, 0.6248
Viterbi labels:  ['O','LOC','LOC'] score = 5.80
P(y*|x) = e^5.8/Z = 330.30/800.598 = 0.4126
```

PART III

Factor Graphs and Message Passing

Make the factorization explicit, then let the factors talk

Table 8.2. One family, many models: the log-linear form $\exp\{\theta^\top f - A(\theta)\}$.

Model	Variables	Sufficient statistics f	Log-partition A
Ising	$x_i \in \{\pm 1\}$	x_i, $x_i x_j$ on edges	#P-hard in general; $\log(4\cosh J)$ for 2 spins
Potts	$x_i \in \{1..q\}$	$\mathbb{1}[x_i{=}s]$, $\mathbb{1}[x_i{=}x_j]$	like Ising
Gaussian MRF	$x_i \in \mathbb{R}$	x_i (weight h_i), $x_i x_j$ (weight $-\frac{1}{2}K_{ij}$)	$\frac{1}{2}\mathbf{h}^\top K^{-1}\mathbf{h} - \frac{1}{2}\log\lvert K\rvert + \frac{n}{2}\log 2\pi$
Linear-chain CRF	labels y_t given $\mathbf{x}$	$f(y_t, \mathbf{x})$, $f(y_{t-1}, y_t)$	$\log Z(\mathbf{x})$ by forward algorithm
RBM	$v_i, h_j \in \{0,1\}$	v_i, h_j, $v_i h_j$	intractable; $P(\mathbf{h} \mid \mathbf{v})$ factorizes

PART III · FACTOR GRAPHS

8.6 Factor Graphs

Definition 8.6 (Factor graph)**.** A *factor graph* for $\tilde{P}(\mathbf{x}) = \prod_{a \in F} f_a(\mathbf{x}_{\partial a})$ is the bipartite graph with one *variable node* per X_i, one *factor node* (drawn as a square) per f_a, and an edge i–a whenever X_i is in the scope ∂a of f_a. We write ∂i for the factors that mention X_i.

A Markov network only records which variables share *some* potential; the factor graph records *which* potentials exist. A triangle, for instance, has (at least) two factor graphs: a single three-way factor (for binary variables a table of $2^3 = 8$ log-potentials, of which $8 - 1 = 7$ are free because adding a constant to all of them is absorbed by Z) and three pairwise factors (in log-linear form: 3 fields plus 3 couplings $= 6$ free parameters). They define genuinely different families.

Worked Example 8.6 (Parity is not pairwise)**.** Let $P(\mathbf{x}) = \frac{1}{4}\mathbb{1}[x_1 \oplus x_2 \oplus x_3 = 0]$ on $\{0,1\}^3$ (even parity). Show that no pairwise model reproduces it, and find the best pairwise approximation in KL divergence.

Solution. It is a single three-way factor $\psi(x_1, x_2, x_3) = \mathbb{1}[x_1 \oplus x_2 \oplus x_3 = 0]$, supported on the four configurations $000, 011, 101, 110$. Every pairwise marginal is uniform: for any (x_i, x_j) exactly one value of the third variable completes even parity, so $P(x_i, x_j) = \frac{1}{4}$; every singleton marginal is $\frac{1}{2}$. A pairwise model $\prod_{i<j}\psi_{ij}(x_i, x_j)/Z$ can vanish at an odd configuration only if some $\psi_{ij}(x_i, x_j) = 0$ there; but every pair value (x_i, x_j) also occurs in an even configuration, which would then get probability 0 too. So no pairwise model equals P. For a pairwise log-linear family Q_θ, $\mathrm{KL}(P\|Q_\theta) = \sum_{\mathbf{x}} P(\mathbf{x})\log P(\mathbf{x}) - \mathbb{E}_P[\theta^\top f] + A(\theta)$, so minimizing it means maximizing $\mathbb{E}_P[\theta^\top f] - A(\theta)$, the log-likelihood (8.10) with P in place of the empirical distribution. By Theorem 8.4 the minimizer matches the singleton and pairwise marginals of P. The uniform distribution (fields and couplings all 0) has exactly these marginals, so it is the best pairwise approximation, and

$$\mathrm{KL}(P\|\text{uniform}) = \sum_{\mathbf{x}:\,P(\mathbf{x})>0} \tfrac{1}{4}\log\frac{1/4}{1/8} = 4 \cdot \tfrac{1}{4}\ln 2 = \ln 2 = 0.6931 \text{ nats}:$$

all three-way information is lost. This is why higher-order factors exist, and why a Markov-network drawing alone under-specifies the model.

PYTHON LAB · s08_examples.py · ex_parity_triangle()

```
pairwise marginals all [0.25 0.25 0.25 0.25]
best pairwise (max-ent) model = uniform; KL(p||q) = 0.6931 nats
```

Converting models. A *Markov network* becomes a factor graph with one factor per potential; a *Bayesian network* with one factor $f_i = P(x_i \mid \mathrm{pa}_i)$ per CPT (no moralization needed; evidence is a unary indicator factor). The factor graph keeps the factorization but forgets directions, so a collider's $D \perp\!\!\!\perp I$ is no longer readable: factor graphs are a language for *computation*, DAGs and undirected graphs for *independence*.

PART III · FACTOR GRAPHS

8.7 Sum-Product on Factor Graphs

Two kinds of message travel along the edges, each a function of the variable on that edge:

$$\mu_{i\to a}(x_i) = \prod_{b\in\partial i\setminus\{a\}} \mu_{b\to i}(x_i), \tag{8.7}$$

$$\mu_{a\to i}(x_i) = \sum_{\mathbf{x}_{\partial a\setminus i}} f_a(\mathbf{x}_{\partial a}) \prod_{j\in\partial a\setminus\{i\}} \mu_{j\to a}(x_j), \tag{8.8}$$

and the belief of a variable is $b_i(x_i) = \prod_{a\in\partial i} \mu_{a\to i}(x_i)$. A variable leaf sends the constant 1; a factor leaf (a unary factor) sends itself. A variable passes on the product of what it heard; a factor multiplies what it heard into its table and sums out everybody except the recipient.

Theorem 8.2 (Sum-product is exact on tree factor graphs)**.** *If the factor graph is a tree, the two-pass schedule (leaves to a root, then back) computes every message once, and $b_i(x_i) = \sum_{\mathbf{x}_{-i}} \tilde{P}(\mathbf{x})$ for every i. Hence $Z = \sum_{x_i} b_i(x_i)$ for any i, and $P(x_i) = b_i(x_i)/Z$. Factor beliefs $b_a(\mathbf{x}_{\partial a}) = f_a \prod_{j\in\partial a} \mu_{j\to a}$ give the clique marginals.*

Proof. Removing an edge i–a splits the tree; let $T_{a\to i}$ be the part containing a. By induction on its size, $\mu_{a\to i}(x_i)$ equals the sum over all variables of $T_{a\to i}$ (other than x_i) of the product of its factors: for a leaf factor this is (8.8) with an empty product; in general $T_{a\to i}$ is a plus the disjoint subtrees hanging off the variables $j \in \partial a \setminus i$, each of which is j plus the subtrees $T_{b\to j}$, $b \in \partial j \setminus a$. The subtrees share no variables, so by the distributive law (Session 7) the sum factorizes into the products (8.7)–(8.8). Since $V \setminus \{i\}$ is the disjoint union of the subtrees $T_{a\to i}$, $a \in \partial i$, the belief is the full marginal. This is the proof of Session 7's tree theorem, with factors as nodes in their own right. □

With d states per variable, all marginals cost $O\big(\sum_a |\partial a|\, d^{|\partial a|}\big)$, linear in the size of the factor tables; on a pairwise tree, a variable-to-factor message followed by a factor-to-variable message is exactly Session 7's message $m_{i\to j}$.

Worked Example 8.7 (Messages on a chain, by hand)**.** Factor graph $f_A - X_1 - f_B - X_2 - f_C - X_3 - f_D$ with binary variables $X_i \in \{0, 1\}$; vectors list the values at 0 and 1: $f_A(x_1) = (1, 3)$, $f_D(x_3) = (2, 1)$, and $f_B(x_1, x_2) = f_C(x_2, x_3) = \left(\begin{smallmatrix} 4 & 1 \\ 1 & 4 \end{smallmatrix}\right)$ (rows = first argument; value 4 when the two variables agree, 1 otherwise). Compute Z and all marginals.

Solution. 1 Leaves: $\mu_{A\to1}=f_A=(1,3)$, $\mu_{D\to3}=f_D=(2,1)$; each variable has only one other factor, so it forwards what it received, $\mu_{1\to B}=(1,3)$, $\mu_{3\to C}=(2,1)$. 2 Factors to X_2 by (8.8): $\mu_{B\to2}(x_2)=\sum_{x_1}f_B(x_1,x_2)\mu_{1\to B}(x_1)=(4\cdot1+1\cdot3,\,1\cdot1+4\cdot3)=(7,13)$ and $\mu_{C\to2}(x_2)=\sum_{x_3}f_C(x_2,x_3)\mu_{3\to C}(x_3)=(4\cdot2+1\cdot1,\,1\cdot2+4\cdot1)=(9,6)$. 3 Belief $b_2=\mu_{B\to2}\mu_{C\to2}=(7\cdot9,\,13\cdot6)=(63,78)$, so $Z=63+78=141$ and $P(X_2{=}1)=78/141=0.5532$. 4 Back-pass: X_2 sends to f_B what it heard from f_C, $\mu_{2\to B}=\mu_{C\to2}=(9,6)$, so $\mu_{B\to1}(x_1)=\sum_{x_2}f_B(x_1,x_2)\mu_{2\to B}(x_2)=(4{\cdot}9{+}1{\cdot}6,\,1{\cdot}9{+}4{\cdot}6)=(42,33)$ and $b_1=\mu_{A\to1}\mu_{B\to1}=(1\cdot42,\,3\cdot33)=(42,99)$, again summing to 141, so $P(X_1{=}1)=99/141=0.7021$. Similarly $\mu_{2\to C}=\mu_{B\to2}=(7,13)$, $\mu_{C\to3}=(4{\cdot}7{+}1{\cdot}13,\,1{\cdot}7{+}4{\cdot}13)=(41,59)$, $b_3=(2{\cdot}41,\,1{\cdot}59)=(82,59)$ and $P(X_3{=}1)=59/141=0.4184$. Ten messages (two per edge, except the two into the leaf factors f_A,f_D, which no marginal needs) give all three marginals and Z; Fig. 8.5 shows the six of the first pass.

PYTHON LAB · s08_examples.py · ex_sum_product_chain()

```
mu_{fB->X2} = [7.  13.]  mu_{fC->X2} = [9.  6.]  belief X2 = [63.  78.]  Z =
141.0
brute-force Z = 141.0
mu_{X2->fB} = [9.  6.]  mu_{fB->X1} = [42.  33.]  mu_{X2->fC} = [7.  13.]
mu_{fC->X3} = [41.  59.]
P(X2=1) = 0.5532; belief X1 = [42.  99.]  P(X1=1) = 0.7021
belief X3 = [82.  59.]  P(X3=1) = 0.4184
max-product:  max-marginal X2 = [32.  48.]
MAP = (1, 1, 1), weight = 48, P(MAP) = 0.3404; max-sum:  [3.4657 3.8712]
```

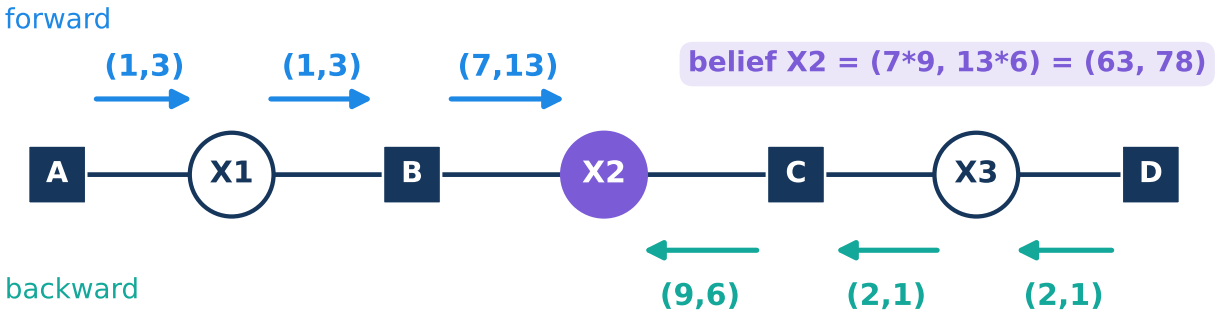


Figure 8.5. Sum-product messages for Worked Example 8.7. Forward (blue) and backward (teal) messages meet at X_2; their product is the unnormalized marginal $(63,78)$, whose sum is $Z=141$.

8.7.1 Max-sum and MAP in factor graphs

Replacing $\sum$ by max in (8.8) gives *max-product*; in logarithms, with $\log f_a$ and sums of messages, *max-sum*. On a tree the max-belief $\max_{\mathbf{x}_{-i}}\tilde{P}(\mathbf{x})$ is obtained at every node, and an MAP assignment follows by fixing the root at its arg-max and back-tracking the stored arg-maxes of each factor (Session 7, max-product VE). In energy language, max-sum is *min-sum*: minimize $E(\mathbf{x})=\sum_a E_a(\mathbf{x}_{\partial a})$.

Worked Example 8.8 (MAP on the same chain)**.** For the factors of Worked Example 8.7, find the MAP configuration and its probability by max-product.

Solution. 1 $\mu^{\max}_{B\to 2}(x_2) = \max_{x_1} f_A(x_1) f_B(x_1, x_2) = (\max(1\cdot 4, 3\cdot 1), \max(1\cdot 1, 3\cdot 4)) = (4, 12)$ and $\mu^{\max}_{C\to 2}(x_2) = \max_{x_3} f_C(x_2, x_3) f_D(x_3) = (\max(4\cdot 2, 1\cdot 1), \max(1\cdot 2, 4\cdot 1)) = (8, 4)$. 2 The max-marginal of X_2 is $(4\cdot 8,\ 12\cdot 4) = (32, 48)$, so $x_2^\star = 1$. 3 Back-tracking: $x_1^\star = \arg\max_{x_1} f_A(x_1) f_B(x_1, 1) = \arg\max(1\cdot 1, 3\cdot 4) = 1$ and $x_3^\star = \arg\max_{x_3} f_C(1, x_3) f_D(x_3) = \arg\max(1\cdot 2, 4\cdot 1) = 1$. The MAP $(1, 1, 1)$ has weight $f_A(1) f_B(1,1) f_C(1,1) f_D(1) = 3\cdot 4\cdot 4\cdot 1 = 48$ (equal to the max-marginal entry 48, as it must be) and probability $48/141 = 0.3404$; in max-sum form the max-marginal is $(\log 32, \log 48) = (3.4657, 3.8712)$ (last line of the `ex_sum_product_chain()` box after Worked Example 8.7). Note that X_3's own marginal favours 0 ($P(X_3{=}1) = 0.4184$) while the MAP sets it to 1.

PART III · FACTOR GRAPHS

8.8 Loopy Belief Propagation

On a factor graph with cycles, initialize all messages to 1 and iterate (8.7)–(8.8) until the messages stop changing, normalizing each message to sum to one (only ratios matter) and, if the iteration oscillates, *damping* it: new message $= (1-\lambda)$ update $+\lambda$ old message, $0 < \lambda < 1$. Convergence is not guaranteed; fixed points are stationary points of the *Bethe free energy* (Koller & Friedman 2009, Ch. 11) and beliefs are approximations, typically over-confident because evidence returns around cycles as if it were new.

Worked Example 8.9 (A 4-cycle Ising model)**.** Ising spins $x_i \in \{\pm 1\}$ on the 4-cycle $X_1{-}X_2{-}X_3{-}X_4{-}X_1$, $P(\mathbf{x}) \propto \exp\{h_1 x_1 + J(x_1x_2 + x_2x_3 + x_3x_4 + x_4x_1)\}$ with $h_1 = 0.2$ and no other fields. Compare loopy BP (pairwise messages started uniform, no damping, 200 parallel updates) with exact enumeration of the $2^4 = 16$ states for $P(X_3{=}+1)$ and $P(X_1{=}+1)$.
Solution. The exact value of $P(X_1{=}+1)$ is the same for every J: flipping x_2, x_3, x_4 together with x_1 leaves the coupling terms unchanged, so $\sum_{x_2,x_3,x_4} e^{J(\cdots)}$ is the same for $x_1 = \pm 1$ and $P(X_1{=}+1) = e^{0.2}/(e^{0.2} + e^{-0.2}) = \sigma(0.4) = 0.5987$. For weak coupling BP agrees well ($J = 0.2$: $P(X_3{=}+1) = 0.5077$ for both, $P(X_1{=}+1) = 0.5990$ versus 0.5987). At $J = 0.5$ BP gives $P(X_3{=}+1) = 0.5440$ versus 0.5403; at $J = 1$ it gives 0.6631 versus 0.5857, and even the directly influenced node X_1 is over-confident (0.6879 versus 0.5987). Figure 8.6 plots the X_3 comparison for J from 0 to 1.6. Removing the edge X_4–X_1 turns the cycle into a chain (a tree), and BP with $J = 1$ becomes exact to machine precision (2.2×10^{-16}).

PYTHON LAB · `s08_examples.py` · `ex_loopy_bp()`

```
J=0.2:  P(X3=+1) loopy BP = 0.5077, exact = 0.5077
P(X1=+1) BP = 0.5990, exact = 0.5987
J=0.5:  P(X3=+1) loopy BP = 0.5440, exact = 0.5403
P(X1=+1) BP = 0.6077, exact = 0.5987
J=1.0:  P(X3=+1) loopy BP = 0.6631, exact = 0.5857
P(X1=+1) BP = 0.6879, exact = 0.5987
chain (tree), J=1:  max |BP - exact| = 2.2e-16
exact P(X1=+1) = sigma(0.4) = 0.5987 for every J
```

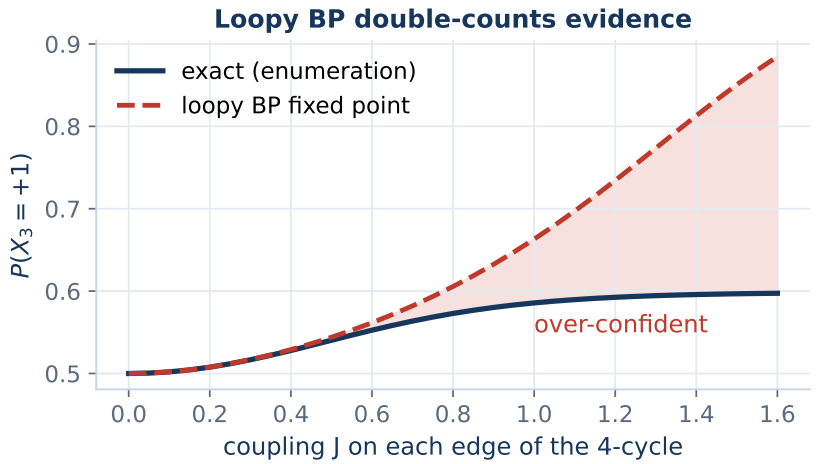


Figure 8.6. $P(X_3{=}+1)$ for the 4-cycle Ising model of Worked Example 8.9 ($h_1 = 0.2$) as a function of the coupling J: loopy BP is exact for $J = 0$, accurate for weak couplings, and increasingly over-confident as the field's evidence circulates around the loop.

PART IV

The Statistics of Undirected Models

Exponential families, moment matching, Gaussian fields and pseudo-likelihood

PART IV · MATHEMATICAL STATISTICS

8.9 Exponential Families and the Log-Partition Function

Theorem 8.3 (Derivatives of the log-partition function)**.** *For the log-linear model* (8.3) *on a finite state space,*

$$\nabla A(\theta) = \mathbb{E}_\theta[f(\mathbf{X})], \qquad \nabla^2 A(\theta) = \operatorname{Cov}_\theta[f(\mathbf{X})]. \tag{8.9}$$

Hence A is convex, and strictly convex if the representation is minimal *(no non-zero $\mathbf{a}$ with $\mathbf{a}^\top f(\mathbf{x})$ constant in $\mathbf{x}$).*

Proof. Write $Z(\theta) = \sum_{\mathbf{x}} e^{\theta^\top f(\mathbf{x})}$, a finite sum of smooth functions. Then

$$\frac{\partial A}{\partial \theta_k} = \frac{1}{Z}\sum_{\mathbf{x}} f_k(\mathbf{x}) e^{\theta^\top f(\mathbf{x})} = \sum_{\mathbf{x}} f_k(\mathbf{x}) P_\theta(\mathbf{x}) = \mathbb{E}_\theta[f_k].$$

Differentiating again, using $\partial P_\theta(\mathbf{x})/\partial\theta_l = P_\theta(\mathbf{x})\big(f_l(\mathbf{x}) - \mathbb{E}_\theta f_l\big)$,

$$\frac{\partial^2 A}{\partial\theta_k \partial\theta_l} = \sum_{\mathbf{x}} f_k(\mathbf{x}) P_\theta(\mathbf{x})\big(f_l(\mathbf{x}) - \mathbb{E}_\theta f_l\big) = \mathbb{E}_\theta[f_k f_l] - \mathbb{E}_\theta[f_k]\mathbb{E}_\theta[f_l].$$

A covariance matrix is positive semi-definite, so A is convex. If $\mathbf{a}^\top \operatorname{Cov}_\theta[f]\mathbf{a} = \operatorname{Var}_\theta(\mathbf{a}^\top f) = 0$ then, since $P_\theta > 0$ everywhere, $\mathbf{a}^\top f$ is constant, which minimality excludes; so the Hessian is positive definite. □

MATHEMATICAL STATISTICS · The cumulant function generates the moments

- A is the cumulant generating function of f under P_θ: $A(\theta+\mathbf{t})-A(\theta)=\log \mathbb{E}_\theta e^{\mathbf{t}^\top f}$, so all cumulants of f are derivatives of A.
- The map $\theta \mapsto \boldsymbol{\mu}=\nabla A(\theta)$ (natural to mean parameters) is one-to-one for minimal families (strict convexity of A); its range is the interior of the convex hull of $\{f(\mathbf{x})\}$, a set called the *marginal polytope* (Wainwright & Jordan 2008, Thm. 3.3).
- The Fisher information of one observation is $I(\theta)=-\mathbb{E}\,\nabla^2 \log P_\theta=\nabla^2 A(\theta)=\mathrm{Cov}_\theta[f]$.

Numerical check of Theorem 8.3. Take the 3-node Ising chain $X_1-X_2-X_3$ with features $f=(x_1,x_2,x_3,\,x_1x_2,\,x_2x_3)$ and parameters $\theta=(h_1,h_2,h_3,J_{12},J_{23})=(0.3,-0.2,0.1,0.8,-0.5)$. Enumerating the $2^3=8$ states gives $A(\theta)$, $\mathbb{E}_\theta f$ and $\mathrm{Cov}_\theta f$ exactly; central finite differences of A (step 10^{-5}) give ∇A and $\nabla^2 A$. They agree to 2.1×10^{-11} (gradient) and 1.6×10^{-6} (Hessian, limited by the finite-difference error), and the smallest eigenvalue of $\mathrm{Cov}_\theta f$ is $0.2274>0$, so A is strictly convex here.

PYTHON LAB · s08_examples.py · ex_expfamily_derivatives()

```
3-node Ising chain, theta = (0.3,-0.2,0.1,0.8,-0.5):  A = 2.5208
E[f] = [0.1375 -0.0501 0.1015 0.6228 -0.4624]
max |grad A - E f| = 2.1e-11, max |Hess A - Cov f| = 1.6e-06
min eigenvalue of Cov[f] = 0.2274 > 0
```

MATHEMATICAL STATISTICS · Why Z is hard: partition functions count

If all potentials are 0/1 clause indicators, Z counts the satisfying assignments of a SAT instance, a #P-complete problem: for $(x_1\vee x_2)(\neg x_2\vee x_3)(\neg x_1\vee\neg x_3)$, $Z=2$, from $(0,1,1)$ and $(1,0,0)$: if $x_1=x_2=0$ the first clause fails; if $x_2=1$ the second clause forces $x_3=1$ and the third then forces $x_1=0$, giving $(0,1,1)$; if $x_1=1,x_2=0$ the third clause forces $x_3=0$, giving $(1,0,0)$ (`ex_sharp_p_counting()`). Z of a general Ising model is #P-hard too; exact computation needs small treewidth (Session 7), otherwise MCMC or variational bounds (Session 11). A 64×64 binary image has $2^{4096}=10^{4096\log_{10}2}\approx10^{1233}$ states.

PART IV · MATHEMATICAL STATISTICS

8.10 Maximum Likelihood is Moment Matching

Theorem 8.4 (MLE in log-linear models)**.** *Let $\mathbf{x}^{(1)},\dots,\mathbf{x}^{(N)}$ be i.i.d. from (8.3) and $\hat{\boldsymbol{\mu}}=\frac{1}{N}\sum_{m=1}^{N} f(\mathbf{x}^{(m)})$ (we index samples by m, since n is the number of variables). The average log-likelihood*

$$\ell_N(\theta)=\theta^\top\hat{\boldsymbol{\mu}}-A(\theta) \tag{8.10}$$

*is concave, with gradient $\hat{\boldsymbol{\mu}}-\mathbb{E}_\theta[f]$. Hence $\hat\theta$ is an MLE if and only if $\mathbb{E}_{\hat\theta}[f]=\hat{\boldsymbol{\mu}}$ (*moment matching*); in a minimal family it is unique when it exists, and it exists if and only if $\hat{\boldsymbol{\mu}}$ lies in the interior of the marginal polytope.*

Proof. $\log P_\theta(\mathbf{x})=\theta^\top f(\mathbf{x})-A(\theta)$; averaging gives (8.10), which depends on the data only through $\hat{\boldsymbol{\mu}}$ (a sufficient statistic). By Theorem 8.3, $\nabla\ell_N=\hat{\boldsymbol{\mu}}-\mathbb{E}_\theta f$ and $\nabla^2\ell_N=-\mathrm{Cov}_\theta f\preceq 0$. A differentiable

concave function is maximized exactly at its stationary points, which are the moment-matching solutions; strict concavity in a minimal family gives uniqueness. Existence: if $\hat{\boldsymbol{\mu}}$ is interior it equals $\nabla A(\theta)$ for some θ (the range of ∇A is the interior of the marginal polytope; second bullet of the mathstat box after Theorem 8.3). If it is on the boundary, there is $\mathbf{a} \neq 0$ with $\mathbf{a}^\top f(\mathbf{x}) \leq \mathbf{a}^\top \hat{\boldsymbol{\mu}}$ for all $\mathbf{x}$ and strict inequality for some $\mathbf{x}$, and then $\ell_N(\theta + t\mathbf{a})$ is strictly increasing in t, so no maximizer exists (e.g. two spins that always agree: $\overline{x_1 x_2} = 1$, take $a = 1$, and $\ell_N(J) = J - \log(4\cosh J)$ increases forever, "$\hat{J} = \infty$"). □

By standard likelihood theory (Casella & Berger 2002; Wasserman 2004), a minimal log-linear MLE is consistent and $\sqrt{N}(\hat{\theta} - \theta) \xrightarrow{d} N\big(0, I(\theta)^{-1}\big)$ with $I(\theta) = \mathrm{Cov}_\theta[f]$. For the two-spin model of Worked Example 8.2, moment matching $\tanh \hat{J} = \overline{x_1 x_2}$ (the sample mean of $x_1 x_2$) gives $\hat{J} = \operatorname{artanh}\big(\overline{x_1 x_2}\big)$, and $I(J) = A''(J) = 1 - \tanh^2 J$.

Worked Example 8.10 (Fitting a 4-cycle Ising model)**.** Draw $N = 2000$ exact samples from the 4-cycle Ising model with

$$\theta = (h_1, \ldots, h_4; J_{12}, J_{23}, J_{34}, J_{41}) = (0.2, -0.1, 0, 0.3;\ 0.5, 0.4, -0.3, 0.6)$$

($x_i \in \{\pm 1\}$, $f = (x_1, \ldots, x_4;\ x_1x_2, x_2x_3, x_3x_4, x_4x_1)$) and fit by Newton's method, $\theta \leftarrow \theta + \mathrm{Cov}_\theta[f]^{-1}(\hat{\boldsymbol{\mu}} - \mathbb{E}_\theta f)$ (the Newton step for (8.10), whose gradient and negative Hessian are $\hat{\boldsymbol{\mu}} - \mathbb{E}_\theta f$ and $\mathrm{Cov}_\theta f$).

Solution. This is a simulation: `ex_mle_moment_matching()` draws the $N = 2000$ samples by enumerating the $2^4 = 16$ states and sampling from their exact probabilities (seed 11). The data moments are $\hat{\boldsymbol{\mu}} = (0.274, -0.001, -0.120, 0.380;\ 0.375, 0.307, -0.214, 0.494)$. Starting from $\theta = 0$, Newton stops after 6 steps (largest step below 10^{-10}) at

$$\hat{\theta} = (0.178, -0.100, -0.034, 0.324;\ 0.482, 0.374, -0.278, 0.542).$$

The asymptotic standard errors $\sqrt{[I(\theta)^{-1}]_{kk}/N}$, with $I(\theta) = \mathrm{Cov}_\theta f$ at the true θ, lie between 0.025 and 0.030; the standardized errors $(\hat{\theta}_k - \theta_k)/\mathrm{se}_k$ are $(-0.73, 0.00, -1.32, 0.86;\ -0.66, -1.05, 0.86, -2.05)$, the largest being 2.05 standard errors (J_{41}: $(0.542 - 0.6)/0.028$, computed from unrounded values), unremarkable among eight coordinates. At $\hat{\theta}$, model moments equal data moments to 1.8×10^{-14}. Every Newton step needed $\mathbb{E}_\theta f$ and $\mathrm{Cov}_\theta f$ — an inference problem; here 2^4 states were enumerated, on a grid one needs BP or MCMC inside the optimizer.

PYTHON LAB · s08_examples.py · ex_mle_moment_matching()

```
empirical moments mu_hat = [0.274 -0.001 -0.12 0.38
0.375 0.307 -0.214 0.494]
Newton steps from theta = 0:  6
MLE = [0.178 -0.1 -0.034 0.324 0.482 0.374 -0.278 0.542]
moment matching:  max |E_mle[f] - mean f| = 1.8e-14
asymptotic s.e.  of MLE (from Fisher info = Cov[f]) =
[0.03 0.028 0.026 0.028 0.027 0.025 0.026 0.028]
standardized errors (MLE - true)/s.e.  = [-0.73 0.  -1.32 0.86 -0.66 -1.05
0.86 -2.05]
max = 2.05
```

8.10.1 Gaussian MRFs: zeros of the precision matrix

Proposition 8.4 (Precision zeros are conditional independences)**.** *Let* $\mathbf{X} \sim N(\boldsymbol{\mu}, \Sigma)$ *with* $K = \Sigma^{-1}$*. For* $i \neq j$*:* $X_i \perp\!\!\!\perp X_j \mid \mathbf{X}_{V\setminus\{i,j\}}$ *if and only if* $K_{ij} = 0$*. Consequently the Gaussian is a Markov network on the graph with edges* $\{(i,j) : K_{ij} \neq 0\}$*, and (being positive) satisfies all three Markov properties for it.*

Proof. Let $a = \{i, j\}$, $b = V \setminus a$. By Gaussian conditioning in precision form (Session 7), $\mathbf{X}_a \mid \mathbf{x}_b$ is bivariate normal with covariance $(K_{aa})^{-1}$, where $K_{aa} = \begin{pmatrix} K_{ii} & K_{ij} \\ K_{ij} & K_{jj} \end{pmatrix}$. Two jointly normal variables are independent iff their covariance vanishes, and the off-diagonal entry of $(K_{aa})^{-1}$ is $-K_{ij}/\det K_{aa}$, which is zero iff $K_{ij} = 0$. The conditional covariance does not depend on $\mathbf{x}_b$, so the statement holds for all values of the conditioning variables. The graph statement follows from (8.5) and Propositions 8.1 and 8.2. □

MATHEMATICAL STATISTICS · Gaussian graphical models revisit Session 5

Session 5 met $(\Sigma^{-1})_{ij} = 0$ as the Gaussian form of conditional independence; now it *is* the graph. The partial correlation $\rho_{ij\cdot\text{rest}} = -K_{ij}/\sqrt{K_{ii}K_{jj}}$ is estimated by inverting the sample covariance (when $N > n$) or, in high dimension, by an ℓ_1-penalized likelihood $\log\det K - \operatorname{tr}(SK) - \lambda\sum_{i\neq j}|K_{ij}|$ (the graphical lasso), which performs structure learning for Gaussian MRFs — a preview of Session 11.

PART IV · MATHEMATICAL STATISTICS

8.11 Pseudo-Likelihood

The MLE needs $A(\theta)$ and its gradient, i.e. inference, at every step. Besag (1975) proposed replacing the joint likelihood by the product of full conditionals, which contain no partition function:

$$\mathrm{PL}_N(\theta) = \frac{1}{N}\sum_{m=1}^{N}\sum_{i=1}^{n}\log P_\theta\big(x_i^{(m)} \mid \mathbf{x}_{-i}^{(m)}\big), \qquad P_\theta(x_i \mid \mathbf{x}_{-i}) = \frac{\exp\{\theta^\top f(x_i, \mathbf{x}_{-i})\}}{\sum_{x_i'}\exp\{\theta^\top f(x_i', \mathbf{x}_{-i})\}}. \tag{8.11}$$

Here $f(x_i', \mathbf{x}_{-i})$ is the feature vector evaluated at the configuration with x_i replaced by x_i'. Each conditional normalizes over the d values of one variable only and involves only the factors touching X_i (all other terms cancel between numerator and denominator). For the Ising model it is the logistic regression of Worked Example 8.3 of each spin on its neighbours, with parameters shared across the n regressions.

Proposition 8.5 (Fisher consistency of pseudo-likelihood)**.** *Let data come from* P_{θ^*} *with* $P_{\theta^*} > 0$*, and let* $\mathrm{PL}(\theta) = \sum_i \mathbb{E}_{\theta^*}\log P_\theta(X_i \mid \mathbf{X}_{-i})$ *be the population objective. Then* θ^* *maximizes* PL*. If the family is minimal and positive,* θ^* *is the unique maximizer.*

Proof. Fix i and $\mathbf{x}_{-i}$. By Gibbs' inequality ($\mathrm{KL} \geq 0$),

$$\mathbb{E}_{\theta^*}\big[\log P_\theta(X_i \mid \mathbf{x}_{-i}) \mid \mathbf{x}_{-i}\big] \leq \mathbb{E}_{\theta^*}\big[\log P_{\theta^*}(X_i \mid \mathbf{x}_{-i}) \mid \mathbf{x}_{-i}\big],$$

with equality iff the two conditionals coincide. Averaging over $\mathbf{x}_{-i}$ and summing over i gives $\mathrm{PL}(\theta) \leq \mathrm{PL}(\theta^*)$. If equality holds, all full conditionals of P_θ and P_{θ^*} agree on every $\mathbf{x}_{-i}$ (all have

positive probability). For positive distributions the full conditionals determine the joint (Brook's lemma: $P(\mathbf{x})/P(\mathbf{x}^0) = \prod_i P(x_i \mid x_{1:i-1}, x^0_{i+1:n})/P(x^0_i \mid x_{1:i-1}, x^0_{i+1:n})$), so $P_\theta = P_{\theta^*}$, and minimality gives $\theta = \theta^*$. □

Because each term of (8.11) is concave in θ (it is the log-likelihood of a log-linear family over the single variable x_i, concave by Theorem 8.3), PL_N is concave and converges pointwise to PL by the law of large numbers; for concave objectives pointwise convergence implies convergence of the maximizers, so the maximum pseudo-likelihood estimator $\hat\theta_{\mathrm{PL}}$ is *consistent*. Standard M-estimation theory (Besag 1975; Koller & Friedman 2009, Ch. 20) gives $\sqrt{N}(\hat\theta_{\mathrm{PL}} - \theta^*) \xrightarrow{d} N(0, H^{-1} V H^{-1})$, where, with $\mathrm{pl}_\theta(\mathbf{x}) = \sum_i \log P_\theta(x_i \mid \mathbf{x}_{-i})$ the contribution of one sample, $H = -\mathbb{E}_{\theta^*} \nabla^2 \mathrm{pl}_{\theta^*}(\mathbf{X})$ and $V = \mathrm{Cov}_{\theta^*} \nabla \mathrm{pl}_{\theta^*}(\mathbf{X})$. This "sandwich" covariance is in general larger than the MLE's $I(\theta^*)^{-1}$ (for the MLE, $H = V = I$).

Simulation. Figure 8.7 compares the two estimators on the 4-cycle Ising model of Worked Example 8.10. For each $N \in \{100, 200, 500, 1000, 2000, 5000, 10000\}$, `ex_mle_vs_pl_consistency()` draws 60 independent data sets of size N exactly from the model (enumeration of the 16 states, seed 12), fits the MLE (Newton) and PL (BFGS on $-\mathrm{PL}_N$), and reports $\mathrm{RMSE} = \left(\frac{1}{60}\sum_{r=1}^{60} \|\hat\theta_r - \theta\|^2\right)^{1/2}$. The reference curve is the Cramér–Rao value $\sqrt{\mathrm{tr}\, I(\theta)^{-1}/N}$ with $\mathrm{tr}\, I(\theta)^{-1} = 12.0294$ computed exactly, e.g. $\sqrt{12.0294/100} = 0.3468$. These simulation outputs give RMSE 0.3601 (MLE) and 0.3626 (PL) at $N = 100$, and 0.0327 and 0.0331 at $N = 10000$ (theory 0.0347). On the $N = 2000$ data set of Worked Example 8.10, PL gives $\hat\theta_{\mathrm{PL}} = (0.177, -0.101, -0.035, 0.324; 0.484, 0.373, -0.276, 0.544)$, within 0.0019 of the MLE in every coordinate.

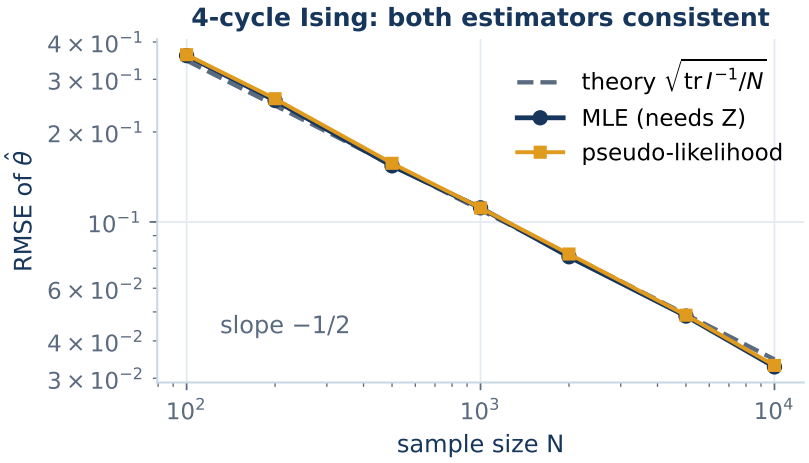


Figure 8.7. Simulated RMSE of MLE and pseudo-likelihood for the 4-cycle Ising model of Worked Example 8.10 (60 replications per N, `ex_mle_vs_pl_consistency()`). Both decay like $N^{-1/2}$ and track the Cramér–Rao rate $\sqrt{\mathrm{tr}\, I^{-1}/N}$; the efficiency loss of PL is small here.

PYTHON LAB · `s08_examples.py` · `ex_mle_vs_pl_consistency()`, `ex_mle_moment_matching()`

```
tr I(theta)^-1 = 12.0294
N=100:  RMSE MLE = 0.3601, PL = 0.3626, theory 0.3468
N=10000:  MLE = 0.0327, PL = 0.0331, theory 0.0347
N=2000 fit:  PL = [0.177 -0.101 -0.035 0.324 0.484 0.373 -0.276 0.544]
max |MLE - PL| = 0.0019
```

PART IV · MATHEMATICAL STATISTICS

8.12 Worked Examples

Worked Example 8.11 (Moralizing and reading independence)**.** Moralize the student network and verify $S \perp\!\!\!\perp L \mid I, G$ using ordinary separation.
Solution. The moral graph adds edge D–I (marrying G's parents) and keeps D–G, I–G, I–S, G–L (Fig. 8.3). Removing $\{I, G\}$ from the graph: S's only edge was to I (now removed), so S becomes isolated; L's only edge was to G (also removed), so L becomes isolated too. S and L land in different (trivial, single-node) components, so $S \perp_H L \mid \{I, G\}$. This matches the DAG's own d-separation answer (the only path $S \leftarrow I \rightarrow G \rightarrow L$ is blocked at the observed non-colliders I and G), as it must: by Propositions 8.1 and 8.3 every separation in the moral graph is an independence of the BN distribution. The converse fails: the DAG's marginal independence $D \perp\!\!\!\perp I$ cannot be read from the moral graph, which contains the edge D–I.

Worked Example 8.12 (A 4-cycle with no DAG equivalent)**.** Four collaborating scientists A, B, C, D arranged in a review cycle (A reviews B's work, B reviews C's, C reviews D's, D reviews A's) each subtly bias their own opinions toward whichever colleague reviewed them, but only directly adjacent scientists influence each other. Model this as a Markov network with edges A–B, B–C, C–D, D–A and pairwise potentials $\phi(x_i, x_j)$ favoring agreement. By separation, $A \perp C \mid \{B, D\}$ and $B \perp D \mid \{A, C\}$. Attempt to draw a DAG with the same four nodes reproducing exactly these independences. Such a DAG must have exactly the cycle's four edges: adjacent pairs are dependent given every conditioning set, so they must be adjacent, while A, C and B, D are independent given some set, so they must not be (a DAG edge makes its endpoints dependent given everything). Now any acyclic assignment of directions to the 4-cycle's edges creates at least one node with two incoming edges, from its two cycle neighbours, which are not adjacent to each other: an unshielded collider. Say C is a collider of $B \rightarrow C \leftarrow D$; the required statement $B \perp\!\!\!\perp D \mid \{A, C\}$ conditions on C, which by Session 6 opens the path $B \rightarrow C \leftarrow D$, so B and D would be d-connected — a contradiction (a collider at A is handled identically; a collider at B or D, whose parents are A and C, contradicts $A \perp\!\!\!\perp C \mid \{B, D\}$ in the same way). Of the $2^4 = 16$ orientations, 14 are acyclic (the two excluded ones are the directed cycles $A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$ and its reverse) and every one of them has at least one collider. (Adding a chord to remove the collider destroys one of the two independences instead.) This confirms the model families are genuinely different, not just different notations for the same thing.

PYTHON LAB · s08_examples.py · ex_four_cycle_orientations()

```
acyclic orientations:  14 of 16; minimum number of colliders:  1
```

Worked Example 8.13 (Computing with Gibbs potentials via VE)**.** For the 2×2 image-denoising grid with pixels $X_{11}, X_{12}, X_{21}, X_{22} \in \{0, 1\}$ and pairwise potentials $\phi(x_i, x_j) = e^{x_i x_j}$ (encouraging agreement) on each of the four grid edges, compute Z by variable elimination: eliminate X_{22} first (appears in two potentials, with X_{12} and X_{21}), multiply those two potentials and sum over $X_{22} \in \{0, 1\}$ to get a new factor $\tau(X_{12}, X_{21})$; continue eliminating X_{21}, then X_{12}, then X_{11}, multiplying whatever factors remain at each step.
Solution. The grid edges are X_{11}–X_{12}, X_{11}–X_{21}, X_{12}–X_{22}, X_{21}–X_{22}; tables below list rows by

the first argument $(0,1)$ and columns by the second. (1) $\tau_1(x_{12},x_{21}) = \sum_{x_{22}} e^{x_{12}x_{22}}e^{x_{21}x_{22}} = e^0 + e^{x_{12}+x_{21}} = 1 + e^{x_{12}+x_{21}}$, i.e. $\tau_1 = \left(\begin{smallmatrix} 2 & 3.7183 \\ 3.7183 & 8.3891 \end{smallmatrix}\right)$ $(1+e^0,\ 1+e^1,\ 1+e^2)$. (2) Eliminating X_{21} (potentials $\phi(x_{11},x_{21})$ and τ_1): $\tau_2(x_{11},x_{12}) = \sum_{x_{21}} e^{x_{11}x_{21}}\tau_1(x_{12},x_{21})$; for $x_{11}=0$ both weights are 1, giving $\tau_2(0,0) = 2 + 3.7183 = 5.7183$ and $\tau_2(0,1) = 3.7183 + 8.3891 = 12.1073$; for $x_{11}=1$ the $x_{21}=1$ term is multiplied by e, giving $\tau_2(1,0) = 2 + e\cdot 3.7183 = 12.1073$ and $\tau_2(1,1) = 3.7183 + e\cdot 8.3891 = 26.5221$. (3) Eliminating X_{12} (potentials $\phi(x_{11},x_{12})$ and τ_2): $\tau_3(x_{11}) = \sum_{x_{12}} e^{x_{11}x_{12}}\tau_2(x_{11},x_{12})$, so $\tau_3(0) = 5.7183 + 12.1073 = 17.8256$ and $\tau_3(1) = 12.1073 + e\cdot 26.5221 = 84.2019$. (4) Finally $Z = \sum_{x_{11}} \tau_3(x_{11}) = 17.8256 + 84.2019 = 102.0275$, the same as the brute-force sum over $2^4 = 16$ configurations. For example, all four pixels equal to 1 makes each of the four potentials e^1, so $P(\text{all four} = 1) = e^4/Z = 54.5982/102.0275 = 0.5351$. The mechanics are identical, term for term, to Session 7's VE on a Bayesian network — only the interpretation of the factors (compatibility scores rather than CPTs) and the presence of an explicit Z to normalize at the very end have changed.

PYTHON LAB · s08_examples.py · ex_grid_partition_ve()

```
tau1(x12,x21) = {(0,0):  2.0, (0,1):  3.7183,
(1,0):  3.7183, (1,1):  8.3891}
tau2(x11,x12) = {(0,0):  5.7183, (0,1):  12.1073,
(1,0):  12.1073, (1,1):  26.5221}
tau3(x11) = {0:  17.8256, 1:  84.2019}; Z (VE) = 102.0275
Z (brute force) = 102.0275
P(all four pixels = 1) = e^4/Z = 54.5982/102.0275 = 0.5351
```

Additional Worked Examples (Instructor Supplement)

Worked Example 8.14 (Computing an actual Gibbs distribution, start to finish)**.** Let $X_1, X_2 \in \{0,1\}$ with a single pairwise potential $\phi(x_1,x_2) = e^{x_1x_2}$ (only the joint configuration $(1,1)$ is rewarded). Compute the full normalized distribution.

Solution. $\phi(0,0) = \phi(0,1) = \phi(1,0) = e^0 = 1$, and $\phi(1,1) = e^1 \approx 2.71828$. The partition function is $Z = 1+1+1+2.71828 = 5.71828$. Normalizing,

$$P(1,1) = \frac{2.71828}{5.71828} \approx \mathbf{0.4754}, \qquad P(0,0) = P(0,1) = P(1,0) = \frac{1}{5.71828} \approx \mathbf{0.1749} \text{ each,}$$

which indeed sum to 1 ($0.4754 + 3(0.1749) = 0.4754 + 0.5247 = 1.0001$, i.e. 1 up to rounding; exactly, $(2.71828+3)/5.71828 = 1$). This is the full mechanics Definition 8.1 describes only symbolically: a raw compatibility score for each configuration, entirely un-normalized on its own, becomes a genuine probability distribution only after dividing by Z — and note that the four potential values sum to $Z = 5.71828$, not to 1: unlike the entries of a CPT, they need not sum to 1 over anything.

Worked Example 8.15 (Numerically confirming why moralization must marry the parents of a collider)**.** Consider $D \to G \leftarrow I$ with D, I binary, independent, $P(D{=}1) = P(I{=}1) = 0.5$ (so each pair (d,i) has $P(d,i) = 0.5\cdot 0.5 = 0.25$) and $P(G{=}1 \mid D, I)$ given by $P(G{=}1 \mid 0,0){=}0.1$, $P(G{=}1 \mid 0,1){=}0.9$, $P(G{=}1 \mid 1,0){=}0.9$, $P(G{=}1 \mid 1,1){=}0.95$. If moralization *skipped* marrying

D, I, undirected separation would (incorrectly) claim $D \perp I \mid G$. Check this numerically.

$$P(D{=}1, I{=}1, G{=}1) = 0.25(0.95) = 0.2375, \quad P(D{=}1, I{=}0, G{=}1) = 0.25(0.9) = 0.225,$$

$$P(D{=}0, I{=}1, G{=}1) = 0.25(0.9) = 0.225, \quad P(D{=}0, I{=}0, G{=}1) = 0.25(0.1) = 0.025,$$

so $P(G{=}1) = 0.2375 + 0.225 + 0.225 + 0.025 = 0.7125$. Then $P(D{=}1 \mid G{=}1) = (0.2375 + 0.225)/0.7125 = 0.4625/0.7125 \approx 0.6491$, and by the symmetry of this particular CPT ($P(G{=}1 \mid 0, 1) = P(G{=}1 \mid 1, 0)$), $P(I{=}1 \mid G{=}1) = (0.2375 + 0.225)/0.7125 \approx 0.6491$ too. If $D \perp I \mid G$ held, we would need $P(D{=}1, I{=}1 \mid G{=}1) = 0.6491 \times 0.6491 \approx 0.4214$. But the true value is $P(D{=}1, I{=}1 \mid G{=}1) = 0.2375/0.7125 \approx \mathbf{0.3333}$ — nowhere near 0.4214. The gap (0.3333 vs. 0.4214) is exactly the numerical fingerprint of explaining-away, and it is precisely why the moral graph *must* include the married edge D–I: without it, ordinary graph separation would confidently predict a wrong number, not just a qualitatively wrong independence claim.

PYTHON LAB · s08_examples.py · ex_gibbs_two_node(), ex_moralization_check()

```
phi(1, 1) = 2.71828 P(1, 1) = 0.4754; P(0,0)=P(0,1)=P(1,0) = 0.1749
Z = 5.71828
P(G=1) = 0.7125; P(D=1|G=1) = 0.6491; product (if D indep I | G) = 0.4214
true P(D=1,I=1|G=1) = 0.3333
```

PART V

Undirected Models in Modern AI

Energies, perception, agent pipelines, coordination and factored RL

PART V · AI CONNECTIONS

8.13 Energy-Based Generative Models

An *energy-based model* is a Gibbs distribution $p_\theta(\mathbf{x}) = e^{-E_\theta(\mathbf{x})}/Z(\theta)$ whose energy is a neural network. The *restricted Boltzmann machine* (RBM) is the classic case: a bipartite pairwise MRF between visible $\mathbf{v} \in \{0,1\}^m$ and hidden $\mathbf{h} \in \{0,1\}^k$ with $E(\mathbf{v}, \mathbf{h}) = -\mathbf{a}^\top \mathbf{v} - \mathbf{b}^\top \mathbf{h} - \mathbf{v}^\top W \mathbf{h}$.

Worked Example 8.16 (RBM conditionals factorize)**.** Show that $P(\mathbf{h} \mid \mathbf{v}) = \prod_j P(h_j \mid \mathbf{v})$ with $P(h_j{=}1 \mid \mathbf{v}) = \sigma(b_j + \mathbf{v}^\top W_{:j})$ ($W_{:j}$ is column j of the 3×2 matrix W), and evaluate it and the free energy for the hypothetical RBM with $\mathbf{v} = (0, 1, 1)$, $\mathbf{a} = (0.1, -0.2, 0)$, $\mathbf{b} = (-1, 0.3)$, $W = \begin{pmatrix} 0.5 & -0.4 \\ -1 & 0.8 \\ 1.5 & -0.2 \end{pmatrix}$.

Solution. Separation: hidden units have no edges among themselves, so $\mathbf{v}$ separates them (Proposition 8.1). Algebraically, with $c_j = b_j + \mathbf{v}^\top W_{:j}$, $e^{-E(\mathbf{v},\mathbf{h})} = e^{\mathbf{a}^\top \mathbf{v}} \prod_j e^{h_j c_j}$, a product over j; dividing by $\sum_{\mathbf{h}} e^{-E} = e^{\mathbf{a}^\top \mathbf{v}} \prod_j (1 + e^{c_j})$ gives $P(\mathbf{h} \mid \mathbf{v}) = \prod_j e^{h_j c_j}/(1 + e^{c_j})$, so $P(h_j{=}1 \mid \mathbf{v}) = e^{c_j}/(1 + e^{c_j}) = \sigma(c_j)$. Numerically, $\mathbf{v}^\top W$ is the sum of rows 2 and 3 of W, $(-1 + 1.5, 0.8 - 0.2) = (0.5, 0.6)$, so $\mathbf{c} = \mathbf{b} + \mathbf{v}^\top W = (-0.5, 0.9)$ and $P(h_1{=}1 \mid \mathbf{v}) = \sigma(-0.5) = 0.3775$, $P(h_2{=}1 \mid \mathbf{v}) = \sigma(0.9) = 0.7109$. Summing out $\mathbf{h}$ gives the *free energy*

$F(\mathbf{v}) = -\log \sum_{\mathbf{h}} e^{-E(\mathbf{v},\mathbf{h})} = -\mathbf{a}^\top \mathbf{v} - \sum_j \log(1 + e^{c_j})$. Here $\mathbf{a}^\top \mathbf{v} = -0.2$, $\log(1 + e^{-0.5}) = \log 1.60653 = 0.47408$, $\log(1 + e^{0.9}) = \log 3.45960 = 1.24115$, so $F(\mathbf{v}) = 0.2 - (0.47408 + 1.24115) = 0.2 - 1.71523 = -1.5152$. $p(\mathbf{v}) \propto e^{-F(\mathbf{v})}$, but Z remains intractable, which is why RBMs are trained with sampling-based gradients.

PYTHON LAB · s08_examples.py · ex_rbm_conditional()

```
v^T W = [0.5 0.6]; pre-activations b + v W = [-0.5 0.9]
P(h=1|v) = [0.3775 0.7109]
1 + e^c_j = [1.60653 3.4596]
a^T v = -0.2; log(1+e^c_j) = [0.47408 1.24115] sum = 1.71523
F = 0.2 - 1.71523 = -1.51523
free energy F(v) = -1.5152
```

LINK TO COURSE SESSION 4 · Generative Models

EBMs avoid Z in two ways. Score matching and diffusion models learn $\nabla_{\mathbf{x}} \log p(\mathbf{x}) = -\nabla_{\mathbf{x}} E(\mathbf{x})$, in which Z disappears because it does not depend on $\mathbf{x}$; contrastive divergence approximates $\mathbb{E}_\theta[\nabla E]$ by short Gibbs runs, using exactly the factorized conditionals of Worked Example 8.16. The MLE gradient "data moments minus model moments" of Theorem 8.4 is the same in all of them.

8.13.1 Perception and language tools for agents

AI CONNECTION · Robotics and Autonomous Agents

A robot's occupancy map $\mathbf{x} \in \{\pm 1\}^{48\times 48}$ (illustrative: a ring and a bar) is seen through a sensor that flips each cell independently with probability $p = 0.15$, giving the reading $\mathbf{y}$. The likelihood of one cell is $P(y_i \mid x_i) = 1 - p$ if $y_i = x_i$ and p otherwise, which is proportional to $e^{\eta x_i y_i}$ with $e^{2\eta} = \frac{1-p}{p}$, i.e. $\eta = \frac{1}{2}\ln\frac{1-p}{p} = \frac{1}{2}\ln\frac{0.85}{0.15} = \frac{1}{2}\ln 5.6667 = 0.8673$. With an Ising prior of coupling J the posterior is again an Ising model, $P(\mathbf{x} \mid \mathbf{y}) \propto e^{-E(\mathbf{x})}$ with $E(\mathbf{x}) = -J\sum_{(i,j)} x_i x_j - \eta \sum_i x_i y_i$. Iterated conditional modes (ICM: starting from $\mathbf{x} = \mathbf{y}$, visit the cells in raster order and set each to the sign of its local field $J\sum_{j\in N(i)} x_j + \eta y_i$; a greedy MAP search) is run by `ex_denoising()` with $J = 1$ on one simulated noisy map (seed 8). In this simulation the third sweep changes no cell, so ICM stops after 3 sweeps, and the pixel error falls from 13.24% (noisy map) to 1.87% (Fig. 8.8). Sampling each cell from $\sigma(2 \cdot \text{local field})$ instead of taking its sign is the Gibbs sampler of Worked Example 8.3.

PYTHON LAB · s08_examples.py · ex_denoising()

```
eta = 0.5 ln((1-p)/p) = 0.8673 for flip prob p = 0.15
pixel error:  noisy = 0.1324, ICM (J=1.0) = 0.0187, sweeps = 3
```

AI CONNECTION · Agentic AI

An LLM travel agent must fill slots (`city`, `date`) from a tool's text output before the next function call. A linear-chain CRF (Worked Example 8.5) is cheap, calibrated and auditable: besides the Viterbi labelling (O, LOC, LOC) it returns its probability 0.4126 and the per-token LOC marginals (0.2338, 0.8179, 0.6248). A sensible policy acts on slots whose marginal

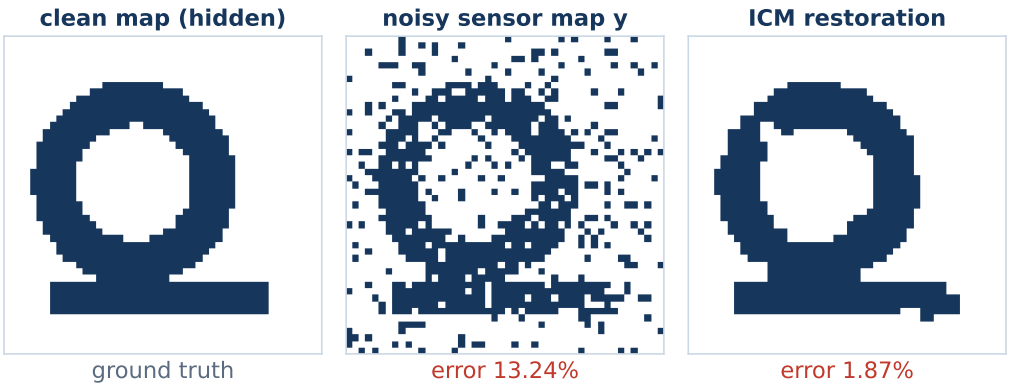


Figure 8.8. MAP denoising of a hypothetical 48×48 occupancy map with an Ising prior ($J = 1$, flip probability $p = 0.15$): hidden truth, noisy sensor map (error 13.24%), ICM restoration (error 1.87%).

exceeds a threshold, say 0.75, and asks the user otherwise: "Paris" ($0.8179 \geq 0.75$) is accepted as the city, "hotel" ($0.6248 < 0.75$) is queried.

LINK TO COURSE SESSION 26 · Agentic AI

Function calling needs structured outputs; an MRF over output fields enforces consistency and exposes uncertainty the planner can act on.

PART V · AI CONNECTIONS

8.14 Coordination Graphs: Multi-Agent Max-Sum

A team of agents with joint action $\mathbf{a} = (a_1, \ldots, a_n)$ often has a payoff that decomposes over a *coordination graph*, $Q(\mathbf{a}) = \sum_{(i,j)\in\mathcal{E}} Q_{ij}(a_i, a_j)$. Maximizing over $|A|^n$ joint actions is exponential; but $\exp Q(\mathbf{a})$ is a pairwise MRF and the maximization is max-sum on its factor graph, each agent sending messages only to its neighbours.

Worked Example 8.17 (Three agents on a line)**.** Agents 1–2–3 with binary actions, $Q_{12} = \left(\begin{smallmatrix} 4 & 1 \\ 0 & 3 \end{smallmatrix}\right)$ (rows a_1), $Q_{23} = \left(\begin{smallmatrix} 1 & 2 \\ 5 & 0 \end{smallmatrix}\right)$ (rows a_2). Find the optimal joint action by max-sum.
Solution. 1 $m_{1\to2}(a_2) = \max_{a_1} Q_{12}(a_1, a_2) = (4, 3)$; 2 $m_{3\to2}(a_2) = \max_{a_3} Q_{23}(a_2, a_3) = (2, 5)$; 3 agent 2 adds them: $(6, 8)$, so $a_2^\star = 1$; back-tracking, $a_1^\star = \arg\max_{a_1} Q_{12}(a_1, 1) = 1$ and $a_3^\star = \arg\max_{a_3} Q_{23}(1, a_3) = 0$, with $Q^\star = 3{+}5 = 8$, the brute-force optimum over all 8 joint actions (Fig. 8.9). An agent 1 acting alone would pick the action holding the largest entry of Q_{12}, $a_1 = 0$ (payoff 4), after which the team can reach at most $\max_{a_2,a_3}[Q_{12}(0, a_2) + Q_{23}(a_2, a_3)] = \max(4 + 2,\ 1 + 5) = 6$: coordination requires the messages.

PYTHON LAB · `s08_examples.py` · `ex_coordination_graph()`

```
m_{1->2}(a2) = [4 3] m_{3->2}(a2) = [2 5] sum = [6 8]
optimal joint action (1, 1, 0), Q* = 8
agent 1 acting alone picks a1 = 0; best value then = 6
```

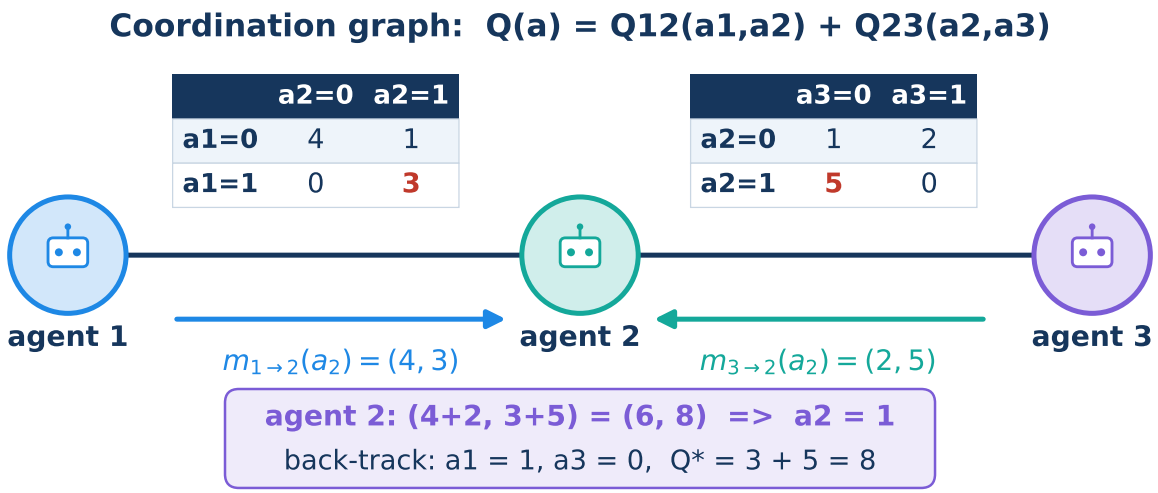


Figure 8.9. A coordination graph as a factor graph: each agent only exchanges a two-entry message with its neighbour, yet the team finds the jointly optimal action.

LINK TO COURSE SESSION 27 · Multi-Agent AI

Coordination graphs make team decisions scale with the number of *interactions*, not joint actions. On cyclic graphs max-sum is loopy and may oscillate (anytime variants keep the best joint action seen). Each message is also a negotiation move: "how good each of your options is for my side of the team".

AI CONNECTION · Reinforcement Learning

In cooperative multi-agent RL and factored MDPs (course session 20), $Q(s,\mathbf{a}) \approx \sum_{(i,j)} Q_{ij}(s,a_i,a_j)$ and the transition model is a dynamic Bayesian network (DBN: a BN repeated over time slices). Greedy action selection is max-sum; a Boltzmann (softmax) policy $\pi(\mathbf{a} \mid s) \propto \exp\{Q(s,\mathbf{a})/\tau\}$ with temperature $\tau > 0$ is *literally* a pairwise MRF over the agents' actions, with marginals from sum-product. For the payoffs of Worked Example 8.17 with $\tau = 1$, the potentials are $e^{Q_{12}} = \left(\begin{smallmatrix} 54.5982 & 2.7183 \\ 1 & 20.0855 \end{smallmatrix}\right)$ and $e^{Q_{23}} = \left(\begin{smallmatrix} 2.7183 & 7.3891 \\ 148.4132 & 1 \end{smallmatrix}\right)$. The messages into agent 2 are the column sums of $e^{Q_{12}}$, (55.5982, 22.8038), and the row sums of $e^{Q_{23}}$, (10.1073, 149.4132); their product is $b_2 =$ (561.95, 3407.19), so $Z = 3969.14$ and $\pi(a_2{=}1) = 3407.19/3969.14 = 0.8584$. Similarly $\pi(a_1{=}1) = \sum_{a_2} e^{Q_{12}(1,a_2)}\,(10.1073, 149.4132)_{a_2}/Z = (10.1073 + 20.0855 \cdot 149.4132)/Z = 0.7586$ and $\pi(a_3{=}0) = \sum_{a_2} (55.5982, 22.8038)_{a_2} e^{Q_{23}(a_2,0)}/Z = 0.8908$. The optimal joint action $(1,1,0)$ has probability $e^{Q^\star}/Z = e^8/Z = 2980.96/3969.14 = 0.7510$; at $\tau = 0.5$ (potentials e^{2Q}) it rises to $e^{16}/Z = 0.9623$.

PYTHON LAB · s08_examples.py · ex_boltzmann_policy(), ex_boltzmann_policy(tau=0.5)

```
m_{1->2}(a2) = [55.5982 22.8038] m_{3->2}(a2) = [10.1073 149.4132]
b2 = [561.95 3407.19]
tau = 1.0:  Z = 3969.14; exp(Q*/tau) = exp(8) = 2980.96
P(a1=1) = 0.7586, P(a2=1) = 0.8584, P(a3=0) = 0.8908
P(optimal joint action (1,1,0)) = 0.7510
tau = 0.5:  Z = 9234119.37; exp(Q*/tau) = exp(16) = 8886110.52
P(optimal joint action (1,1,0)) = 0.9623
```

LINK TO COURSE SESSION 20 · Deep and Model-based RL

Factored world models, value functions and Boltzmann exploration inherit today's tools: the factorization sets the cost of planning, message passing is the inner loop.

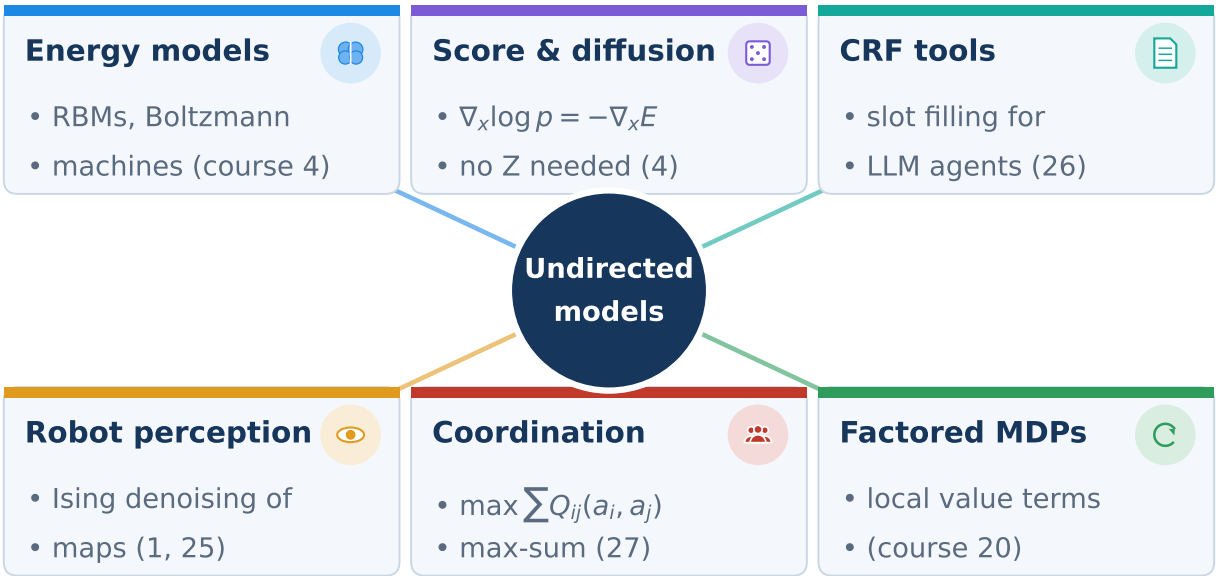


Figure 8.10. Where undirected models and factor graphs reappear in the Advanced AI course (numbers in parentheses are course sessions).

Figure 8.10 and Table 8.3 collect where undirected models reappear in the course.

Table 8.3. This session in the Advanced AI course. "Session k" refers to these notes, "course session m" to the course schedule.

Concept today	Where it reappears	Role there
Gibbs / energy form	generative models (course session 4): EBMs, diffusion	$p \propto e^{-E}$, score $= -\nabla E$
Separation, Markov blanket	Gibbs sampling (Session 11), RBMs	local updates, factorized conditionals
Log-linear MLE	learning PGMs (Session 10), deep nets (course session 3)	moment matching, cross-entropy
CRFs	NLP layers of agentic systems (course session 26)	structured, calibrated outputs
Max-sum on factor graphs	multi-agent coordination (course session 27)	distributed joint-action selection
Boltzmann policies, factored Q	RL (course sessions 17–20)	exploration, factored MDPs
Moral graphs	causal DAGs (Sessions 12–14)	a causal DAG's distribution is a Markov network on its moral graph (Proposition 8.3), so today's inference tools apply

PART V · AI CONNECTIONS

8.15 Summary

THINK ABOUT IT A colleague draws a triangle as a Markov network and says "so the model is pairwise". What information is missing, and which drawing would settle it?

THINK ABOUT IT Why can pseudo-likelihood ignore Z while maximum likelihood cannot — and what does it pay for that?

THINK ABOUT IT In a team of agents on a ring, loopy max-sum may oscillate. Which human negotiation failure does this resemble, and how would you break the loop?

Table 8.4. Cheat sheet of key formulas.

Item	Formula
Gibbs distribution	$P(\mathbf{x}) = \frac{1}{Z}\prod_C \phi_C(\mathbf{x}_C)$, $Z = \sum_{\mathbf{x}} \prod_C \phi_C$; energy $E = -\sum_C \log \phi_C$
Markov properties	Gibbs $\Rightarrow$ G $\Rightarrow$ L $\Rightarrow$ P; $P > 0$: all equivalent (Hammersley–Clifford)
Ising conditional	$P(x_i{=}{+1} \mid \mathbf{x}_{-i}) = \sigma\big(2(h_i + \sum_{j\in N(i)} J_{ij}x_j)\big)$
Factor-graph messages	$\mu_{i\to a} = \prod_{b\in\partial i\setminus a} \mu_{b\to i}$; $\mu_{a\to i} = \sum_{\mathbf{x}_{\partial a\setminus i}} f_a \prod_{j\in\partial a\setminus i} \mu_{j\to a}$
Log-partition	$\nabla A = \mathbb{E}_\theta f$, $\nabla^2 A = \mathrm{Cov}_\theta f = I(\theta)$
MLE	$\mathbb{E}_{\hat\theta} f = \frac{1}{N}\sum_m f(\mathbf{x}^{(m)})$; $\sqrt{N}(\hat\theta - \theta) \xrightarrow{d} N(0, I^{-1})$
Gaussian MRF	$K_{ij} = 0 \iff X_i \perp\!\!\!\perp X_j \mid \text{rest}$; $X_i \mid \text{rest} \sim N(\mu_i - \sum_j \frac{K_{ij}}{K_{ii}}(x_j - \mu_j), \frac{1}{K_{ii}})$
Pseudo-likelihood	$\sum_m \sum_i \log P_\theta(x_i^{(m)} \mid \mathbf{x}_{-i}^{(m)})$: no Z, consistent

Table 8.5. Common pitfalls and how to avoid them.

Pitfall	Remedy
Reading potentials as probabilities	only ratios matter; normalize by Z at the very end
Dropping arrows without marrying parents	moralize first; a collider's parents must be joined
Assuming Markov properties without positivity	check zeros: hard constraints can break (P)$\Rightarrow$(G) and Hammersley–Clifford
Treating a Markov network drawing as "pairwise"	draw the factor graph; triangles may carry three-way factors
Trusting loopy BP like exact BP	check on small cases, damp, beware strong couplings
Fitting an MRF without inference	MLE gradients need model moments; use BP/MCMC or pseudo-likelihood

Further reading. Koller & Friedman (2009), Chs. 4, 11 and 20; Bishop (2006), Secs. 8.3–8.4; Murphy (2022, 2023) on exponential families, undirected models and energy-based models;

Wasserman (2004), Chs. 18–19 on undirected graphs and log-linear models; Wainwright & Jordan (2008) on exponential families, the marginal polytope and the variational view of BP; Besag (1975) for pseudo-likelihood.

APPENDIX

8.16 Appendix: Proof of the Hammersley–Clifford Theorem

The forward direction (Gibbs ⇒ global Markov) is Proposition 8.1: factor the Gibbs distribution into the piece depending on one side of a separator S and the piece depending on the other, exactly mirroring Session 5's factorized-CI form $g(x,z)h(y,z)$ generalized to cliques. For the converse, by Proposition 8.2 it suffices to show that a strictly positive P satisfying the *pairwise* property factorizes over the cliques of H. The argument uses the Möbius inversion formula.

Construction. Fix a reference configuration $\mathbf{x}^0$. For $S \subseteq V$ let $H_S(\mathbf{x}) = \log P(\mathbf{x}_S, \mathbf{x}^0_{V\setminus S})$ (well defined by positivity), and for each $C \subseteq V$ define

$$V_C(\mathbf{x}) = \sum_{S \subseteq C} (-1)^{|C\setminus S|} H_S(\mathbf{x}).$$

Möbius inversion on the lattice of subsets gives $H_A = \sum_{C\subseteq A} V_C$ for every A; with $A = V$, $\log P(\mathbf{x}) = H_V(\mathbf{x}) = \sum_{C\subseteq V} V_C(\mathbf{x})$, and each V_C depends on $\mathbf{x}$ only through $\mathbf{x}_C$.

Key lemma: $V_C \equiv 0$ unless C is a clique. If C contains non-adjacent i, j, group the subsets of C in fours, S, $S\cup\{i\}$, $S\cup\{j\}$, $S\cup\{i,j\}$ with $S \subseteq C\setminus\{i,j\}$:

$$V_C = \sum_{S \subseteq C\setminus\{i,j\}} (-1)^{|C\setminus\{i,j\}\setminus S|} \big[H_{S\cup\{i,j\}} - H_{S\cup\{i\}} - H_{S\cup\{j\}} + H_S\big].$$

With $\mathbf{w}$ the configuration of $V\setminus\{i,j\}$ (equal to $\mathbf{x}$ on S and to $\mathbf{x}^0$ elsewhere), the bracket is

$$\log \frac{P(x_i \mid x_j, \mathbf{w})}{P(x_i^0 \mid x_j, \mathbf{w})} - \log \frac{P(x_i \mid x_j^0, \mathbf{w})}{P(x_i^0 \mid x_j^0, \mathbf{w})} = 0,$$

because the pairwise property $X_i \perp\!\!\!\perp X_j \mid \mathbf{X}_{V\setminus\{i,j\}}$ says the conditional of X_i does not depend on x_j. Hence $P(\mathbf{x}) = \prod_{C \text{ clique}} e^{V_C(\mathbf{x}_C)}$: a Gibbs factorization with $\phi_C = e^{V_C}$ (the constant $V_\emptyset = \log P(\mathbf{x}^0)$ is absorbed into Z). Both positivity (to take logarithms and condition) and the pairwise property are used; Remark 8.1 shows the theorem fails without positivity. See Koller & Friedman (2009), Ch. 4.

CONNECTIONS AND TAKEAWAYS

We now have two complementary graphical languages — directed (Session 6) and undirected (today) — each encoding independence via a different local pattern-matching rule, both computable exactly with the same variable-elimination machinery (Session 7), and unified for computation by factor graphs, on which sum-product and max-sum run unchanged. Between them, essentially any independence structure a domain expert can describe is representable by *some* graphical model.

Every session so far, however, made an unstated assumption: that we already *know* the graph G (or H) and the numerical parameters Θ that go with it. Part IV gave a first taste

of the answer for undirected models — maximum likelihood is moment matching, and it needs inference at every step. **Session 10** tackles the numbers in general: given a fixed graph structure and a dataset, how do we estimate Θ by maximum likelihood (in closed form, for Bayesian networks) or by Bayesian updating — and what happens when some of the variables in our carefully-drawn graph are never actually observed in the data at all?